Supervyse OPF Server Setting Guide

Revision 1.1
08/2023

Copyright (c) 2024, All Rights Reserved. Insyde Software.

No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language or computer language, in any form, or by any means, electronic, mechanical, magnetic, optical, chemical, manual or otherwise, without the prior written permission of Insyde Software Corp.

Disclaimer

Insyde Software provides this document and the programs "as is" without warranty of any kind, either expressed or implied, including, but not limited to, the implied warranties of merchantability or fitness for a particular purpose.

This document could contain technical inaccuracies or typographical errors. Changes are periodically made to the information herein; these changes will be incorporated in future revisions of this document. Insyde Software is under no obligation to notify any person of the changes.

The following trademarks are used in this document:

Insyde and InsydeH2O are registered trademarks or trademarks of Insyde Software. All other trademarks or trade names are property of their respective holders.

1. Supervyse OPF DDNS Server Setting Guide 6
1.1. Agenda 6
1.2. Prerequisites 6
1.2.1. Domain Name Server 6
1.2.1.1. Installation 6
1.2.1.2. Configuration file 6
1.2.1.2.1. Modify Configuration File 7
1.2.1.3. Add Zones/Reverse-Zones to Configuration 7
1.2.1.4. Create zone database 8
1.2.1.5. Generate/Edit the TSIG key information 9
1.2.1.5.1. Add TSIG Key to Configuration File 9
1.2.1.6. Restart the service 9
1.2.1.7. Check the status update on DDNS server 9
1.2.2. DHCP server (Optional) 10
1.2.2.1. Installation 10
1.2.2.2. Configuration file 10
1.2.2.2.1. IPv4 10
1.2.2.3. Restart DHCP server 11
1.2.2.4. Log messages 11
1.3. DDNS function validation 11
1.3.1. DNS Update Without TSIG 11
1.3.2. DNS Update With TSIG 12
1.3.3. DHCP Client FQDN 12
1.3.3.1. IPv4 12
1.4. Supervyse OPF DDNS Configuration 13
1.4.1. DNS Update without TSIG Authentication 13
1.4.2. DNS Update with TSIG Authentication 13
1.4.3. DHCP Client FQDN 14
2. Supervyse OPF RADIUS Server Setting Guide 16
2.1. Introduction 16
2.2. Prerequisites 16
2.3. RADIUS on Ubuntu 20.04 17
2.3.1. FreeRADIUS installation 17
2.3.2. Configure FreeRADIUS 3.0 17
2.4. RADIUS authentication setting in BMC Web Service page 22
2.5. RADIUS authentication setting by IPMI Commands 23
3. Supervyse OPF TACACS+ Server Setting Guide 24
3.1. Introduction 24
3.2. Prerequisites 24
3.3. Ubuntu 20.04 25
3.3.1. Install tacacs+ 25
3.3.2. Configuring your TACACS+ server setting 25
3.3.2.1. Configuring key for encryption 25
3.3.2.2. Configuring User information 27
3.3.2.3. Check setting correct 28
3.3.3. Running TACACS+ Server 28
3.4. TACACS+ authentication setting in BMC Web Service page 29
3.5. TACACS+ authentication setting by IPMI Commands 30
4. Supervyse OPF Active Directory Server Setting Guide 31
4.1. Introduction 31
4.2. Configuring AD service 31
4.3. Configuring User and Group 32

4.4. Configuring Bmcweb Active Directory 34
4.5. Active Directory Login 35
4.6. Reference ..... **Error! Bookmark not defined.**

5. Supervyse OPF LDAP Server Setting Guide 38

5.1. LDAP 38
5.2. LDAP server on ubuntu 38
5.2.1. Environment setup 38
5.2.2. Configuring slapd 39
5.2.3. LDAP Browser/Editor 39
5.2.4. Example for add user and set privilege group 40
5.2.4.1. Setting on LDAP server 40
5.2.4.2. Setting on OPF BMC 46
5.2.4.3. Login Web test 47

6. Supervyse OPF Kerberos SSO Login Setting Guide 48

6.1. Kerberos 48
6.1.1. Introduction 48
6.1.2. Prerequisites 48
6.1.3. Windows server 2016 49
6.1.4. Ubuntu 53
6.1.5. BMC Setup 54
6.1.6. SSO Login 60

Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2023-05-10	-	-	RV23.unicorn Release Document
1.1	2023-08-23	Roy Kuo	Brad Chang	Add RADIUS and TACACS+ server setting guide

1. Supervyse OPF DDNS Server Setting Guide

1.1. Agenda

- This page includes the following sections:
  1. DNS server setting guide
  2. DHCP server setting guide
  3. Validate DDNS update by nsupdate command
  4. Validate DDNS update by DHCP client FQDN method
  5. DDNS Verification from Supervyse SPF WEB Settings

1.2. Prerequisites

- You will need the following packages installed on **Ubuntu** for DDNS function validation.

1.2.1. Domain Name Server

- **D**ynamic **DNS** (DDNS or DynDNS) is a method of automatically updating a name server in the Domain Name System (DNS), often in real time, with the activeDNS configuration of its configured hostnames, addresses or other information.

1.2.1.1. Installation

```
# sudo apt-get install bind9
```

1.2.1.2. Configuration file

- Adding or making sure the line of **"include "/etc/bind/named.conf.local";"** in the configurations file: /etc/bind/named.conf

```
# sudo vi /etc/bind/named.conf
```

```
// This is the primary configuration file for the BIND DNS server named.
//
// Please read /usr/share/doc/bind9/README.Debian.gz for information on the
// structure of BIND configuration files in Debian, *BEFORE* you customize
// this configuration file.
//
// If you are just adding zones, please do that in /etc/bind/named.conf.local

include "/etc/bind/named.conf.options";
include "/etc/bind/named.conf.local";
include "/etc/bind/named.conf.default-zones";

...
```

1.2.1.2.1. Modify Configuration File

- Add the two files listed below in the /etc/bind/named.conf.local
  "/etc/bind/rndc.key": this file includes the TSIG key that is used for DDNS authentication
  "/etc/bind/zones.bmc": this file stores the supportted domain/zone settings

```
# sudo vi /etc/bind/named.conf.local
```

```
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

include "/etc/bind/rndc.key";
include "/etc/bind/zones.bmc";
```

1.2.1.3. Add Zones/Reverse-Zones to Configuration

- Two domain names are going to be created in the DDNS test.
  The domain name of "tsig.io" is used for DDNS with the TSIG authentication
  The domain name of "iscbmc.com" is used for DDNS without the TSIG authentication
  Note 1: Zones allow users to query the IP addresses with domain-name.
  Note 2: Reverse-zones allow users to query the domain name with IP addresses.

```
# sudo vi /etc/bind/zones.bmc
```

```
...

//the Zones(.tsig.io) used for the DDNS test with TSIG Authentication
// zone file: "/var/lib/bind/db.tsig.io"
zone "tsig.io" {
    type master;
    file "/var/lib/bind/db.tsig.io";
    allow-update { key "tsig2"; };
};

//the Zones(.iscbmc.com) used for the DDNS test without TSIG Authentication
// zone file: "/var/lib/bind/db.iscbmc.com"
zone "iscbmc.com" {
    type master;
    file "/var/lib/bind/db.iscbmc.com";
    allow-update { any; };
};

//the Reverse-Zones(.tsig.io) used for the DDNS test with TSIG Authentication
// zone file: "/var/lib/bind/db.192.168"
// Notes: Enable this configuration when running DDNS with TSIG authentication
//zone "168.192.in-addr.arpa" {
//    type master;
//    file "/var/lib/bind/db.192.168";
//    allow-update { key "tsig2"; };
//};

//the Reverse-Zones(.iscbmc.com) used for the DDNS test without TSIG Authentication
// zone file: "/var/lib/bind/db.192.168"
// Notes: Enable this configuration when running DDNS without TSIG authentication
zone "168.192.in-addr.arpa" {
```

```
    type master;
    file "/var/lib/bind/db.192.168";
    allow-update { any; };
};

...
```

1.2.1.4. Create zone database

- Zone database: iscbmc.com

```
# sudo vi /var/lib/bind/db.iscbmc.com
```

```
$TTL 86400      ; 1 day
@               IN SOA  ns.iscbmc.com. root.ns.iscbmc.com. (
                        9          ; serial
                        604800     ; refresh (1 week)
                        86400      ; retry (1 day)
                        2419200    ; expire (4 weeks)
                        86400      ; minimum (1 day)
                        )
;                   NS      ns.iscbmc.com.
@               IN NS  ns.iscbmc.com.
ns                   A       10.1.1.111
```

- Zone database: tsig.io

```
# sudo vi /var/lib/bind/db.tsig.io
```

```
$TTL 86400      ; 1 day
@                   IN SOA  ns1.tsig.io. root.tsig.io. (
                        7          ; serial
                        604800     ; refresh (1 week)
                        86400      ; retry (1 day)
                        2419200    ; expire (4 weeks)
                        86400      ; minimum (1 day)
                        )
;
@                   IN NS   ns1.tsig.io.

ns1                 A       10.128.0.254
```

- Reverse-Zone database for running DDNS without TSIG authentication: iscbmc.com

```
# sudo vi /var/lib/bind/db.192.168
```

Notes: If you are running DDNS with TSIG authentication, please using this database file for the zone of "tsig.io"

```
$TTL 86400      ; 1 day
@               IN SOA  ns.iscbmc.com. root.ns.iscbmc.com. (
                        4          ; serial
                        604800     ; refresh (1 week)
                        86400      ; retry (1 day)
                        2419200    ; expire (4 weeks)
                        86400      ; minimum (1 day)
                        )
;
@               IN NS  ns.iscbmc.com.
111              IN PTR  ns.iscbmc.com.
```

- Reverse-Zone database for running DDNS with TSIG authentication: tsig.io

```
# sudo vi /var/lib/bind/db.192.168
```

Notes: If you are running DDNS without TSIG authentication, please using this database file for the zone of "iscbmc.com"

```
$TTL 86400      ; 1 day
@               IN SOA  ns.tsig.io. root.tsig.io. (
                        6         ; serial
                        604800     ; refresh (1 week)
                        86400      ; retry (1 day)
                        2419200    ; expire (4 weeks)
                        86400      ; minimum (1 day)
                        )
;
@               IN NS  ns.tsig.io
121             IN PTR  ns.tsig.io
```

1.2.1.5. Generate/Edit the TSIG key information

- Using the dnssec-keygen to generate the TSIG key (**Note: Suggested version of bind-9.3.2**)

```
# dnssec-keygen -a HMAC-MD5 -b 64 -n HOST tsig2
Ktsig2.+157+19089
```

```
# ls Ktsig2.*
Ktsig2.+157+19089.key
Ktsig2.+157+19089.private
```

```
# cat Ktsig2.+157+19089.key
tsig2. IN KEY 512 3 157 XvoUy06bgpM=
```

- Or use tsig-keygen commmand to generate the TSIG key

```
# tsig-keygen tisg2
key "tisg2" {
      algorithm "hmac-sha256;
      secret ""aPECzQeNnpkIrsRMQB75poWNHoBosEN5z6MOkvOjb7c=";
};
```

1.2.1.5.1. Add TSIG Key to Configuration File

```
# sudo vi /etc/bind/rndc.key
```

```
...
key "tsig2" {
       algorithm hmac-md5;
       secret "XvoUy06bgpM=";
};
...
```

1.2.1.6. Restart the service

- Restart the service

```
# systemctl restart bind9.service
```

1.2.1.7. Check the status update on DDNS server

- Monitor the log to check the status update on DDNS server

```
# journalctl -u bind9.service
```

```
// When a client update the IP and domain name information to DNS server, the information on syslog may look as below
Dec  8 09:35:05 bmc named[6627]: client 192.168.88.25#50479/key tsig2: signer "tsig2"     approved
Dec  8 09:35:05 bmc named[6627]: client 192.168.88.25#50479/key tsig2: updating zone 'tsig.io/IN': deleting rrset at 'bbtest.tsig.io' A
Dec  8 09:35:05 bmc named[6627]: client 192.168.88.25#50479/key tsig2: updating zone 'tsig.io/IN': adding an RR at 'bbtest.tsig.io' A 192.168.88.25
Dec  8 09:35:05 bmc named[6627]: client 192.168.88.25#50211/key tsig2: signer "tsig2" approved
Dec  8 09:35:05 bmc named[6627]: client 192.168.88.25#50211/key tsig2: updating zone '168.192.in-addr.arpa/IN': deleting rrset at '25.88.168.192.in-addr.arpa' PTR
Dec  8 09:35:05 bmc named[6627]: client 192.168.88.25#50211/key tsig2: updating zone '168.192.in-addr.arpa/IN': adding an RR at '25.88.168.192.in-addr.arpa' PTR bbtest.tsig.io.
```

1.2.2. DHCP server (Optional)

1.2.2.1. Installation

```
# sudo apt-get install isc-dhcp-server
```

1.2.2.2. Configuration file

1.2.2.2.1. IPv4

```
# sudo vi /etc/dhcp/dhcpd.conf
```

```
ddns-update-style interim;
ddns-updates on;
authoritative;
allow unknown-clients;
update-conflict-detection false;

default-lease-time 600;
max-lease-time 7200;

log-facility local7;

subnet 192.168.87.0 netmask 255.255.255.0 {
  range 192.168.87.130 192.168.87.149;
  option domain-name-servers 192.168.87.1;
  option domain-name "insydebmc.com";
  ddns-domainname "insydebmc.com";
  ddns-rev-domainname "in-addr.arpa.";
  option routers 192.168.87.1;
}

key "web" {
        algorithm hmac-md5;
        secret "YH8Yz4s8+XGAq20Ei8cG4w==";
};

zone insydebmc.com. {
  primary 192.168.87.1;
#  key "web";
}

zone 87.168.192.in-addr.arpa. {
```

```
  primary 192.168.87.1;
#  key "web";
}
```

- 192.168.87.1 is the IP of DHCP server
- Remove # in forward and reverse zone areas to enable TSIG authentication
- **option domain-name-servers** points to IP address of Ubuntu, which is also a DNS server

1.2.2.3. Restart DHCP server

```
# sudo service isc-dhcp-server restart
```

1.2.2.4. Log messages

```
# journalctl -u isc-dhcp-server.service
```

1.3. DDNS function validation

- Scope:
  NS Update without TSIG
  NS Update with TSIG
  DHCP with FQDN
- Purpose:
  Manually verify the DNS server is set up properly

1.3.1. DNS Update Without TSIG

- ***Update the domain name of host1.iscbmc.com with IP 192.168.0.1 to DNS server(192.168.88.196)***

```
# nsupdate
> server 192.168.88.196
> zone iscbmc.com
> update delete host1.iscbmc.com A
> update add host1.iscbmc.com 86400 A 192.168.0.100
> send
> quit
```

- **Log Information on DDNS server**

```
# journalctl -u bind9.service
```

```
 Dec  8 10:15:33 bmc named[14400]: client 192.168.88.25#45448: updating zone 'iscbmc.com/IN': deleting rrset at
'host1.iscbmc.com' A
 Dec  8 10:15:33 bmc named[14400]: client 192.168.88.25#45448: updating zone 'iscbmc.com/IN': adding an RR at
'host1.iscbmc.com' A 192.168.0.100
```

- **Using "nslookup" to query the IP addresses**

```
# nslookup host1.iscbmc.com 192.168.88.196
Server:         192.168.88.196
Address:        192.168.88.196#53

Name:   host1.iscbmc.com
Address: 192.168.0.100
```

1.3.2. DNS Update With TSIG

- ***Update the domain name of host1.tsig.io with IP 192.168.11.11 to DNS server(192.168.88.196)***

```
# nsupdate -y tsig2:XvoUy06bgpM=
> server 192.168.88.196
> zone tsig.io
> update delete host1.tsig.io A
> update add host1.tsig.io 86400 A 192.168.11.11
> send
> quit
```

- **Log Information on DDNS server**

```
 // The syslog information on DDNS server may look as below
 Dec  8 09:43:47 bmc named[6627]: client 192.168.88.25#32854/key tsig2: signer "tsig2" approved
 Dec  8 09:43:47 bmc named[6627]: client 192.168.88.25#32854/key tsig2: updating zone 'tsig.io/IN': deleting rrset at
'host1.tsig.io' A
 Dec  8 09:43:47 bmc named[6627]: client 192.168.88.25#32854/key tsig2: updating zone 'tsig.io/IN': adding an RR at
'host1.tsig.io' A 192.168.11.11
```

- **Using "nslookup" to query the IP addresses**

```
// Using nslookup to check if domain name us update to DNS server
# nslookup host1.tsig.io 192.168.88.196
Server:        192.168.88.196
Address:       192.168.88.196#53

Name:   host1.tsig.io
Address: 192.168.11.11
```

1.3.3. DHCP Client FQDN

Notes: You may also use the "Wireshark" to capture the DHCP packet to check the FQDN information.

1.3.3.1. IPv4

- Let's use busybox's **udhcpc** to verify the setting.
- Suppose we are going to register **udhcpc.insydebmc.com**
- Kill udhcpc and restart it with -F option

```
# killall udhcpc; /sbin/udhcpc -i bond0 -x hostname:udhcpc -S -b -t2 -T2 -F udhcpc -p/var/run/dhcp.bond0.pidudhcpc &
```

- Expected log messages

```
dhcpd: Added new forward map from udhcpc.insydebmc.com to 192.168.87.131
dhcpd: Added reverse map from 131.87.168.192.in-addr.arpa. to udhcpc.insydebmc.com
```

- Resolve the host by using nslookup command on Ubuntu

```
# nslookup udhcpc.insydebmc.com 192.168.87.1
Server:        192.168.87.1
Address:   192.168.87.1#53

Name:    udhcpc.insydebmc.com
Address: 192.168.87.131
```

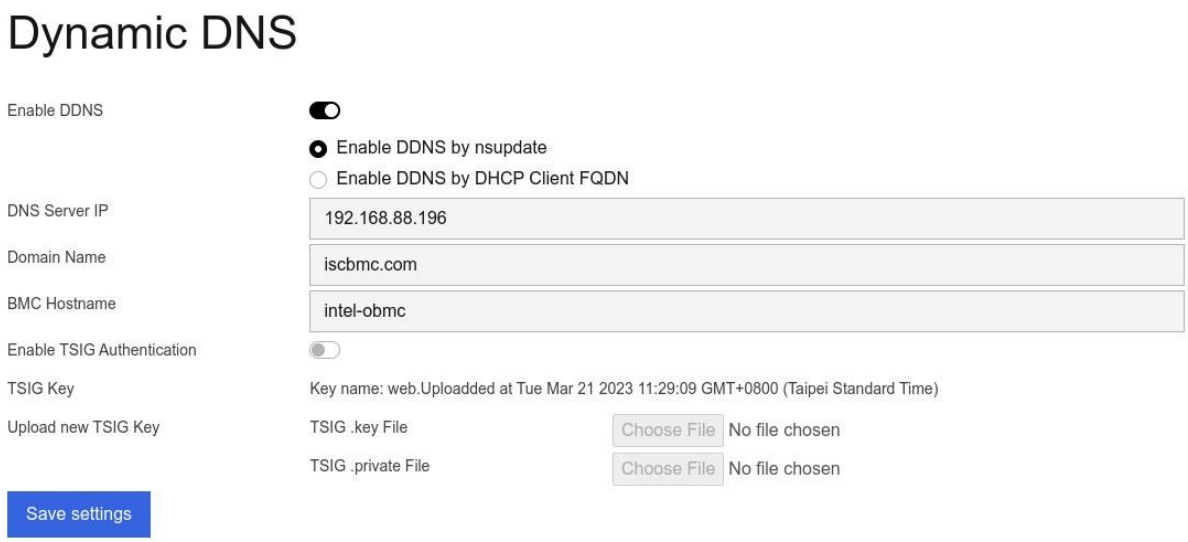
```
# nslookup 192.168.87.131 192.168.87.1
Server:        192.168.87.1
Address:   192.168.87.1#53
```

```
131.87.168.192.in-addr.arpa  name = udhcpc.insydebmc.com.
```

1.4. Supervyse OPF DDNS Configuration

1.4.1. DNS Update without TSIG Authentication

- WEB: Settings -> Dynamic DNS



- Running "nslookup" to query the IP addresses with domain name

```
# nslookup intel-obmc.iscbmc.com 192.168.88.196
Server:       192.168.88.196
Address:   192.168.88.196#53

Name: intel-obmc.iscbmc.com
Address: 192.168.88.25
```

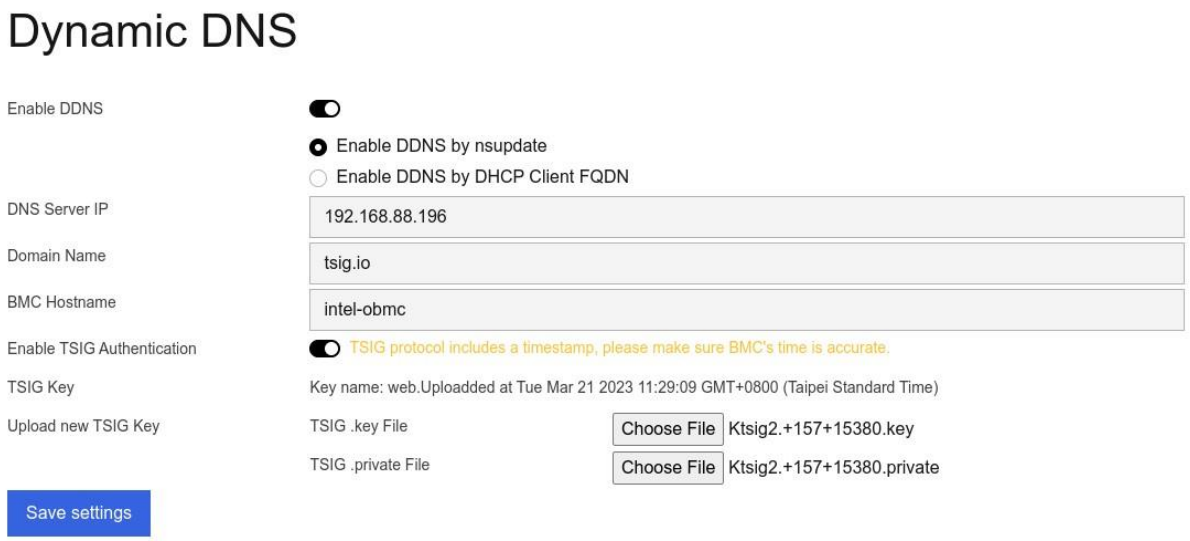
- Running "nslookup" to query the domain name with IP addresses

```
# nslookup 192.168.88.25 192.168.88.196
Server:       192.168.88.196
Address:   192.168.88.196#53

25.88.168.192.in-addr.arpa name = intel-obmc.iscbmc.com.
```

1.4.2. DNS Update with TSIG Authentication

- WEB: Settings -> Dynamic DNS



- Running "nslookup" to query the IP addresses with domain name

```
# nslookup intel-obmc.tsig.io 192.168.88.196
Server:        192.168.88.196
Address:   192.168.88.196#53

Name: intel-obmc.tsig.io
Address: 192.168.88.25
```

- Running "nslookup" to query the domain name with IP addresses

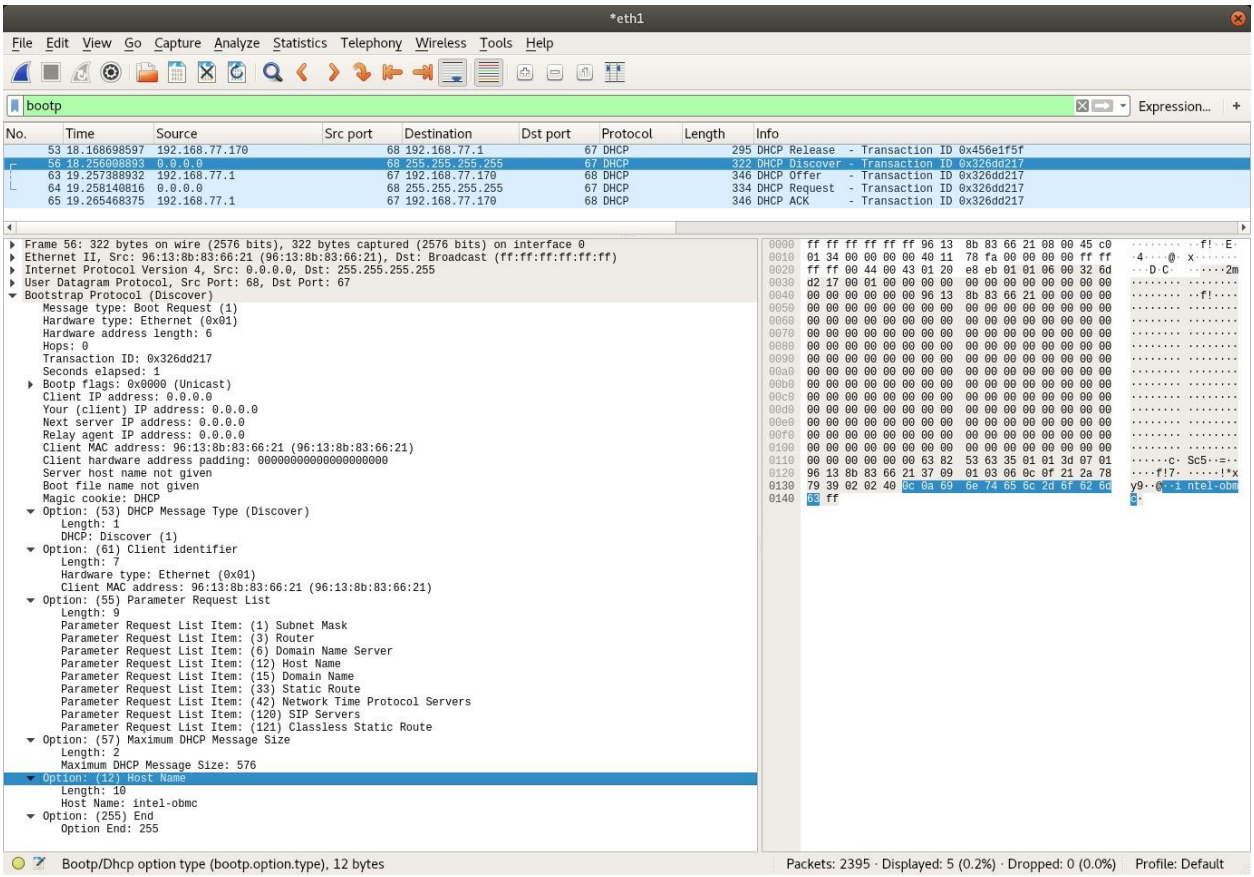
```
# nslookup 192.168.88.25 192.168.88.196
Server:        192.168.88.196
Address:   192.168.88.196#53

25.88.168.192.in-addr.arpa name = intel-obmc.tsig.io.
```

1.4.3. DHCP Client FQDN

- WEB: Settings -> Dynamic DNS

- Run **Wireshark** to capture the DHCP packet, and check the hostname included in the **Host name** field


*eth1
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
bootp
Expression...
No. Time Source Src port Destination Dst port Protocol Length Info
53 18.168698597 192.168.77.170 68 192.168.77.1 67 DHCP 295 DHCP Release - Transaction ID 0x456e1f5f
56 19.250008893 0.0.0.0 68 255.255.255.255 67 DHCP 322 DHCP Discover - Transaction ID 0x326dd217
63 19.257388932 192.168.77.1 67 192.168.77.170 68 DHCP 346 DHCP Offer - Transaction ID 0x326dd217
64 19.258140816 0.0.0.0 68 255.255.255.255 67 DHCP 334 DHCP Request - Transaction ID 0x326dd217
65 19.265468375 192.168.77.1 67 192.168.77.170 68 DHCP 346 DHCP ACK - Transaction ID 0x326dd217
Frame 56: 322 bytes on wire (2576 bits), 322 bytes captured (2576 bits) on interface 0
Ethernet II, Src: 96:13:8b:83:66:21 (96:13:8b:83:66:21), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
User Datagram Protocol, Src Port: 68, Dst Port: 67
Bootstrap Protocol (Discover)
Message type: Boot Request (1)
Hardware type: Ethernet (0x01)
Hardware address length: 6
Hops: 0
Transaction ID: 0x326dd217
Seconds elapsed: 1
Bootp flags: 0x0000 (Unicast)
Client IP address: 0.0.0.0
Your (client) IP address: 0.0.0.0
Next server IP address: 0.0.0.0
Relay agent IP address: 0.0.0.0
Client MAC address: 96:13:8b:83:66:21 (96:13:8b:83:66:21)
Client hardware address padding: 00000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP
Option: (53) DHCP Message Type (Discover)
Length: 1
DHCP: Discover (1)
Option: (61) Client identifier
Length: 7
Hardware type: Ethernet (0x01)
Client MAC address: 96:13:8b:83:66:21 (96:13:8b:83:66:21)
Option: (55) Parameter Request List
Length: 9
Parameter Request List Item: (1) Subnet Mask
Parameter Request List Item: (3) Router
Parameter Request List Item: (6) Domain Name Server
Parameter Request List Item: (12) Host Name
Parameter Request List Item: (15) Domain Name
Parameter Request List Item: (33) Static Route
Parameter Request List Item: (42) Network Time Protocol Servers
Parameter Request List Item: (120) SIP Servers
Parameter Request List Item: (121) Classless Static Route
Option: (57) Maximum DHCP Message Size
Length: 2
Maximum DHCP Message Size: 576
Option: (12) Host Name
Length: 10
Host Name: intel-obmc
Option: (255) End
Option End: 255
Bootp/Dhcp option type (bootp.option.type), 12 bytes
Packets: 2395 · Displayed: 5 (0.2%) · Dropped: 0 (0.0%)
Profile: Default

2. RADIUS Server Setting Guide

2.1. Introduction

- This page is the RADIUS Server setting guide for Ubuntu client and BMC.

2.2. Prerequisites

1. Ensure RADIUS is existed on the BMC firmware image
    - Before building BMC firmware, the following menuconfig option should be enabled, so that the RADIUS authentication is available.
    - OPF could use pkconfig to modify menuconfig, e.g. bitbake pkconfig -c menuconfig

        - **menuconfig -> Security Options -> 3rd Authentication -> Enable RADIUS Authentication**

```
(Top) → Security Options → 3rd Authentication
                                        Main menu
[*] Enable LDAP Authentication
[*]     Enable Active Directory Authentication
[*] Enable RADIUS Authentication
[ ] Enable TACACS Authentication
[ ] Enable Kerberos Authentication
```

        - **menuconfig -> Network Services -> Web Server -> WEB STYLE -> Config WEB Pages -> Security and Access -> RADIUS**

```
(Top) → Network Services → Web Server → WEB STYLE → Config WEB Pages → Security and A
                                                        Main menu
[*] Sessions
[*] Current Users
[*] LDAP
[*] LDAP (Insyde)
[*] Active Directory
[*] RADIUS
[*] TACACS
[*] User management
[*] Users
[*] Policies
[*] Certificates
[*] SSL Certification
[*] Kerberos Certification
[*] Security Settings
[*] IP Access Control
```

    - Choose the menuconfig option to enable the RADIUS authentication for the generated BMC firmware image.

2.3. RADIUS on Ubuntu 20.04

- FreeRADIUS
  includes a RADIUS server, a BSD licensed client library, a PAM library, and an Apache module. In most cases, the word FreeRADIUS refers to the RADIUS server.

2.3.1. FreeRADIUS installation

Install FreeRADIUS and utilities on Ubuntu

```
$ sudo apt install freeradius freeradius-utils
```

Check the FreeRADIUS installation

- OS : Ubuntu 20.04
- FreeRADIUS 3.0.20

```
$ freeradius -v
```

```
insyde@insyde-ubuntu20.04:~$ freeradius -v
radiusd: FreeRADIUS Version 3.0.20, for host x86_64-pc-linux-gnu, built on Jan
5 2020 at 06:11:13
FreeRADIUS Version 3.0.20
```

2.3.2. Configure FreeRADIUS 3.0

Step 1. Add Vendor-Specific attribute to the FreeRADIUS 3.0 dictionary

```
$ sudo nano /etc/freeradius/3.0/dictionary
```

Add the following line:

```
ATTRIBUTE   Vendor-Specific     26  string
```

```
#       See RFC 6158 for more details.
#       http://ietf.org/rfc/rfc6158.txt
#

#
#       These attributes are examples
#
#ATTRIBUTE      My-Local-String         3000    string
#ATTRIBUTE      My-Local-IPAddr         3001    ipaddr
#ATTRIBUTE      My-Local-Integer        3002    integer
ATTRIBUTE       Vendor-Specific         26      string
```

Step 2. Configure RADIUS Client information

Add your client ip, secret and short name:

- secret: **insyde**

```
$ sudo nano /etc/freeradius/3.0/clients.conf
```

Example:
(192.168.0.0/16 here means accept clients with ip starts with "192.168.")

```
client insydebmc {
   ipaddr      = 192.168.0.0/16
   secret      = insyde
}
```

```
#
#  You can now specify one secret for a network of clients.
#  When a client request comes in, the BEST match is chosen.
#  i.e. The entry from the smallest possible network.
#
#client private-network-1 {
#        ipaddr          = 192.0.2.0/24
#        secret          = testing123-1
#}

client insydebmc {
        ipaddr          = 192.168.0.0/16
        secret          = insyde
}

#client private-network-2 {
#        ipaddr          = 198.51.100.0/24
#        secret          = testing123-2
#}
```

Step 3. Configure User information

Add Local users and passwords:

```
$ sudo nano /etc/freeradius/3.0/users
```

Example, we could setting accounts for different privilege:

```
"super"     Cleartext-Password := "super1"
       Vendor-Specific = "H=4, I=4"
"insyde"    Cleartext-Password := "1111"
       Vendor-Specific = "H=3, I=3"
"raduser1"  Cleartext-Password := "123456"
       Vendor-Specific = "H=2, I=2"
```

```
# See raddb/policy.d/filter, filter_username {} section.
#
#"John Doe"      Cleartext-Password := "hello"
#                Reply-Message = "Hello, %{User-Name}"

"super"          Cleartext-Password := "super1"
                 Vendor-Specific = "H=4, I=4"
"insyde"         Cleartext-Password := "1111"
                 Vendor-Specific = "H=3, I=3"
"raduser1"       Cleartext-Password := "123456"
                 Vendor-Specific = "H=2, I=2"


#
# Dial user back and telnet to the default host for that port
#
#Deg     Cleartext-Password := "ge55ged"
#        Service-Type = Callback-Login-User,
```

Please note, BMC uses Vendor-Specific attribution to identify the privilege for login users.
You can set any string as Vendor-Specific attribution with a max length of 64 bytes.
Refer to the example in the table below:

Privilege	Value
Administrator	H=4, I=4
Operator	H=3, I=3
User	H=2, I=2

Step 4. (Optional) Modify FreeRADIUS port

If you do not want to use the default RADIUS port 1812, which is defined in /etc/services, you can change it here:

```
$ sudo nano /etc/freeradius/3.0/sites-available/default
```

```
#               for the host name, only the A record is used.
#
# ipv4addr = *
# ipv6addr = *
ipaddr = *

#  Port on which to listen.
#  Allowed values are:
#        integer port number (1812)
#        0 means "use /etc/services for the proper port"
port = 1812

#  Some systems support binding to an interface, in addition
#  to the IP address.  This feature isn't strictly necessary,
#  but for sites with many IP addresses on one interface,
#  it's useful to say "listen on all addresses for eth0".
#
#  If your system does not support this feature, you will
#  get an error if you try to use it.
#
```

Step 5. Configure FreeRADIUS Authorizing account

In OPF, Authentication and Authorization is separated into two steps. We have to add an Authorizing account (by custom radius script)
Example:

- Authorizing account: **InsydeRadiusAdmin**
- password: **P@ssw0rd**

$ sudo gedit /etc/freeradius/3.0/sites-enabled/default
Add the following script to "authorize" section:

```
if (User-Name == "InsydeRadiusAdmin") {
   if (User-Password =~ /^P@ssw0rd:(.+)$/) {
      # copy original values
      update {
         reply:User-Name := &request:User-Name
         reply:User-Password := &request:User-Password
         request:User-Name := "%{1}"
         request:User-Password := ""
      }
      #
      #  Read the 'users' file.  In v3, this is located in
      #  raddb/mods-config/files/authorize
      files
      # restore original values
      update {
         request:User-Name := &reply:User-Name
         request:User-Password := &reply:User-Password
         reply:User-Name !* ANY
         reply:User-Password !* ANY
      }
      if (&reply:Attr-26) {
         update control {
            Auth-Type := Accept
         }
```

```
    } else {
      update control {
        Auth-Type := Reject
      }
    }
    return
  }
}
```

Open default /etc/freeradius/3.0/sites-enabled

```
282 #  Make *sure* that 'preprocess' comes before any realm if you
283 #  need to setup hints for the remote radius server
284 authorize {
285
286 if (User-Name == "InsydeRadiusAdmin") {
287     if (User-Password =~ /^P@ssw0rd:(.+)$/) {
288         # copy original values
289         update {
290             reply:User-Name := &request:User-Name
291             reply:User-Password := &request:User-Password
292             request:User-Name := "%{1}"
293             request:User-Password := ""
294         }
295         #
296         #  Read the 'users' file.  In v3, this is located in
297         #  raddb/mods-config/files/authorize
298         files
299         # restore original values
300         update {
301             request:User-Name := &reply:User-Name
302             request:User-Password := &reply:User-Password
303             reply:User-Name !* ANY
304             reply:User-Password !* ANY
305         }
306         if (&reply:Attr-26) {
307             update control {
308                 Auth-Type := Accept
309             }
310         } else {
311             update control {
312                 Auth-Type := Reject
313             }
314         }
315         return
316     }
317 }
318
```

authorize section

Step 6. Running RADIUS Server

Restart the RADIUS service to apply settings:

```
$ service freeradius restart
```

To start the server in debugging mode:

```
$ service freeradius stop
$ sudo freeradius -X
```

To start the RADIUS service at boot:

```
$ sudo systemctl enable freeradius
```

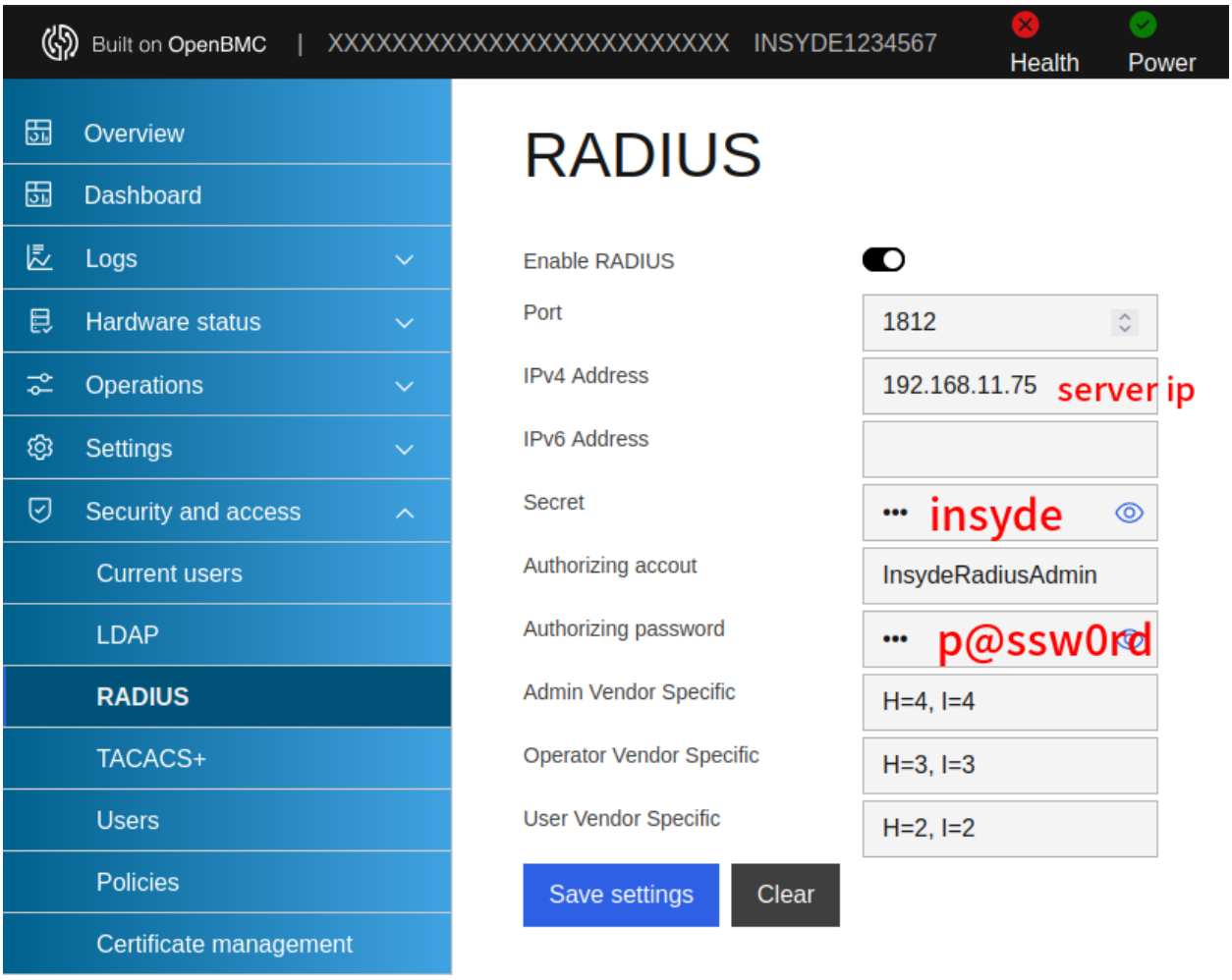
2.4. RADIUS authentication setting in BMC Web Service page

You will need to update the page in this picture

- **IPv4 Address**: ip of your RADIUS server
- **Secret**: "insyde" or your secret configured in the clients.conf
- **Vendor Specific**:
  (Set empty on Vendor Specific will block the valid RADIUS user using the corresponding privilege to login)

Admin Vendor Specific	H=4, I=4
Operator Vendor Specific	H=3, I=3
User Vendor Specific	H=2, I=2

- **Authorizing account**: "InsydeRadiusAdmin"
- **Authorizing passowrd**: "P@ssw0rd"



2.5. RADIUS authentication setting by IPMI Commands

Setting RADIUS in the Supervyse Web Service page will be completed,
for now, here is IPMI Commands: (Modify BMC_IP **192.168.11.209** and server ip **192 168 11 75** for your environment)

```
BMC_IP=192.168.11.209
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x60 1 1
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x60 3 192 168 11 75
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x60 4 `python3 -c "print(' '.join(hex(ord(c)) for c in 'insyde'.ljust(64, '\0')))"`
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x60 5 4 `python3 -c "print(' '.join(hex(ord(c)) for c in 'H=4, I=4'.ljust(64, '\0')))"`
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x60 5 3 `python3 -c "print(' '.join(hex(ord(c)) for c in 'H=3, I=3'.ljust(64, '\0')))"`
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x60 5 2 `python3 -c "print(' '.join(hex(ord(c)) for c in 'H=2, I=2'.ljust(64, '\0')))"`
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x60 7 `python3 -c "print(' '.join(hex(ord(c)) for c in 'InsydeRadiusAdmin'.ljust(64, '\0')))"`
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x60 8 `python3 -c "print(' '.join(hex(ord(c)) for c in 'P@ssw0rd'.ljust(64, '\0')))"`
```

3. TACACS+ Server Setting Guide

3.1. Introduction

- This page is the TACACS+ Server setting guide for Ubuntu client and BMC.

3.2. Prerequisites

1. Ensure TACACS+ is existed on the BMC firmware image
    - Before building BMC firmware, the following menuconfig option should be enabled, so that the TACACS+ authentication is available.
    - OPF could use pkconfig to modify menuconfig, e.g. bitbake pkconfig -c menuconfig

        - **menuconfig -> Security Options -> 3rd Authentication -> Enable TACACS Authentication**

```
(Top) →Security Options →3rd Authentication
                                    Main menu
[ ] Enable LDAP Authentication
[ ] Enable RADIUS Authentication
[*] Enable TACACS Authentication
[ ] Enable Kerberos Authentication
```

        - **menuconfig -> Network Services -> Web Server -> WEB STYLE -> Config WEB Pages -> Security and Access -> TACACS**

```
(Top) →Network Services →Web Server →WEB STYLE →Config WEB Pages →Security and Access
                                                    Main menu
[*] Sessions
[*] Current Users
[*] LDAP
[*] LDAP (Insyde)
[*] Active Directory
[*] RADIUS
[*] TACACS
[*] User management
[*] Users
[*] Policies
[*] Certificates
[*] SSL Certification
[*] Kerberos Certification
[*] Security Settings
[*] IP Access Control
```

            - Choose the menuconfig option to enable the TACACS+ authentication for the generated BMC firmware image.

3.3. Ubuntu 20.04

3.3.1. Install tacacs+

- tacacs+
  includes a TACACS+ server, encrypt password tool and debug mode.

- tacacs+ version : 4.0.4.27a

Use command **sudo apt-get install tacacs+** to install

- Note: Ubuntu 20.04 removed tacacs+ package from official repository, so we require bionic source to install tacacs+.

Ubuntu 18.04 and before

```
# sudo apt-get install tacacs+
```

Ubuntu 20.04 and after

```
# sudo add-apt-repository 'deb http://security.ubuntu.com/ubuntu bionic main universe'
# sudo apt install tacacs+
# sudo add-apt-repository --remove 'deb http://security.ubuntu.com/ubuntu bionic main universe'
# sudo apt update
```

3.3.2. Configuring your TACACS+ server setting

- All TACACS+ server settings are in the file /etc/tacacs+/tac_plus.conf

```
# sudo nano /etc/tacacs+/tac_plus.conf
```

3.3.2.1. Configuring key for encryption

- Specifies an encryption key used to encrypt packets between the server and BMC. This key must match the key configured on the BMC.

- Use command **tac_pwd** to get garbled text (option).

```
  GNU nano 4.8                          /etc/tacacs+/tac_plus.conf
# Created by Henry-Nicolas Tourneur(henry.nicolas@tourneur.be)
# See man(5) tac_plus.conf for more details

# Define where to log accounting data, this is the default.

accounting file = /var/log/tac_plus.acct

# This is the key that clients have to use to access Tacacs+

key = YOUR_KEY

# Use /etc/passwd file to do authentication

#default authentication = file /etc/passwd


user = insyde_1 {
    service = exec {
        priv-lvl = 4
    }
    login = cleartext PASSWD_1
}
```

Example:

```
key = YOUR_KEY
```

3.3.2.2. Configuring User information

Add users, passwords and set privileges.

```
  GNU nano 4.8                                /etc/tacacs+/tac_plus.conf
key = YOUR_KEY

# Use /etc/passwd file to do authentication

#default authentication = file /etc/passwd


user = insyde_1 {
    service = exec {
        priv-lvl = 4
    }
    login = cleartext PASSWD_1
}
user = insyde_2 {
    service = exec {
        priv-lvl = 2
    }
    login = cleartext PASSWD_2
}
user = insyde_3 {
    service = exec {
        priv-lvl = 3
    }
    login = cleartext PASSWD_3
}


# You can use feature like per host key with different enable passwords
#host = 127.0.0.1 {
#        key = test
#        type = cisco
#        enable = <des|cleartext> enablepass
```

Example: Setting accounts for different privileges:

```
user = insyde_1 {
   service = exec {
      priv-lvl = 4
   }
   login = cleartext PASSWD_1
}
user = insyde_2 {
   service = exec {
      priv-lvl = 2
   }
   login = cleartext PASSWD_2
}
user = insyde_3 {
   service = exec {
      priv-lvl = 3
   }
   login = cleartext PASSWD_3
}
```

The table below is the relationship of privilege levels between TACACS+ users and IPMI users:

TACACS+	IPMI
priv-lvl = 2	User
priv-lvl = 3	Operator
priv-lvl = 4	Administrator

3.3.2.3. Check setting correct

Save and quit the edit file. Use the command below to check that the settings are correct.

```
# service tacacs_plus check
```

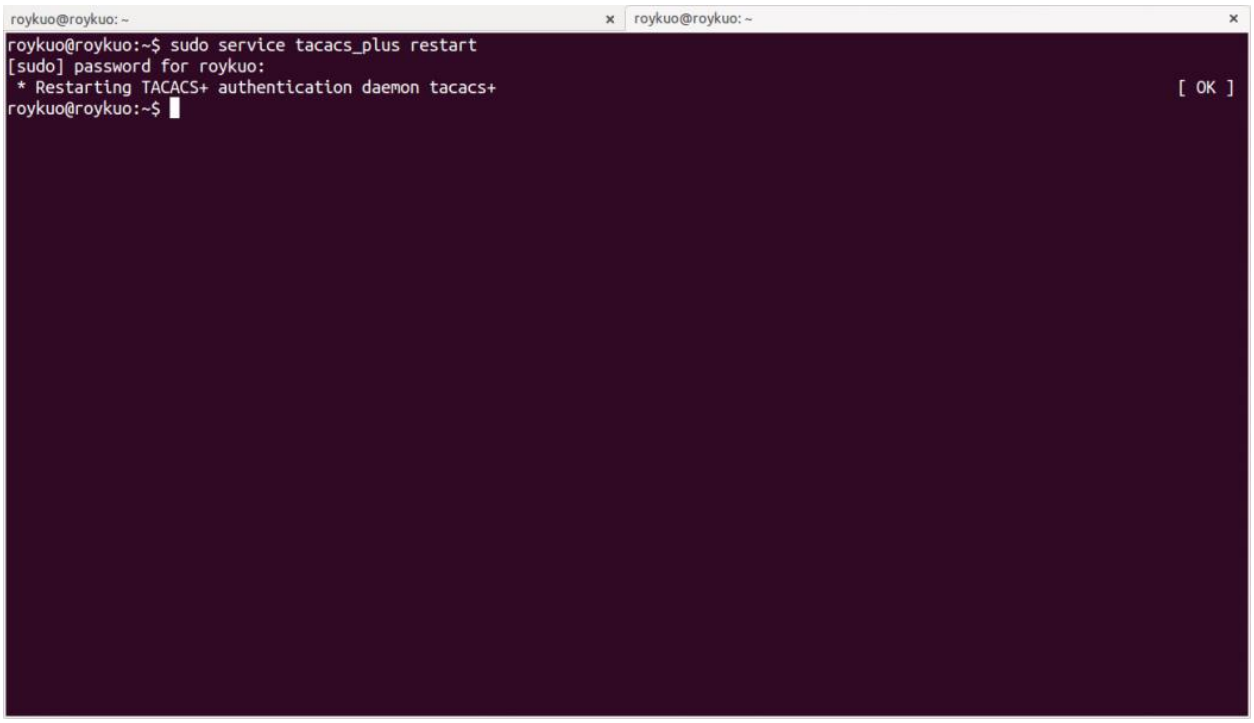
```
# Much more features are availables, like ACL, more service compatibilities,
# commands authorization, scripting authorization.
# See the man page for those features.
user = insyde_1 {
    service = exec {
        priv-lvl = 4
    }
    login = cleartext PASSWD_1
}
user = insyde_2 {
    service = exec {
        priv-lvl = 2
    }
    login = cleartext PASSWD_2
}
user = insyde_3 {
    service = exec {
        priv-lvl = 3
    }
    login = cleartext PASSWD_3
}

 * Checking TACACS+ authentication daemon configuration files successful tacacs+
roy.kuo@SC1-4:~$
```

3.3.3. Running TACACS+ Server

To restart the server service:

```
# sudo service tacacs_plus restart
```



To start the server in debugging mode:

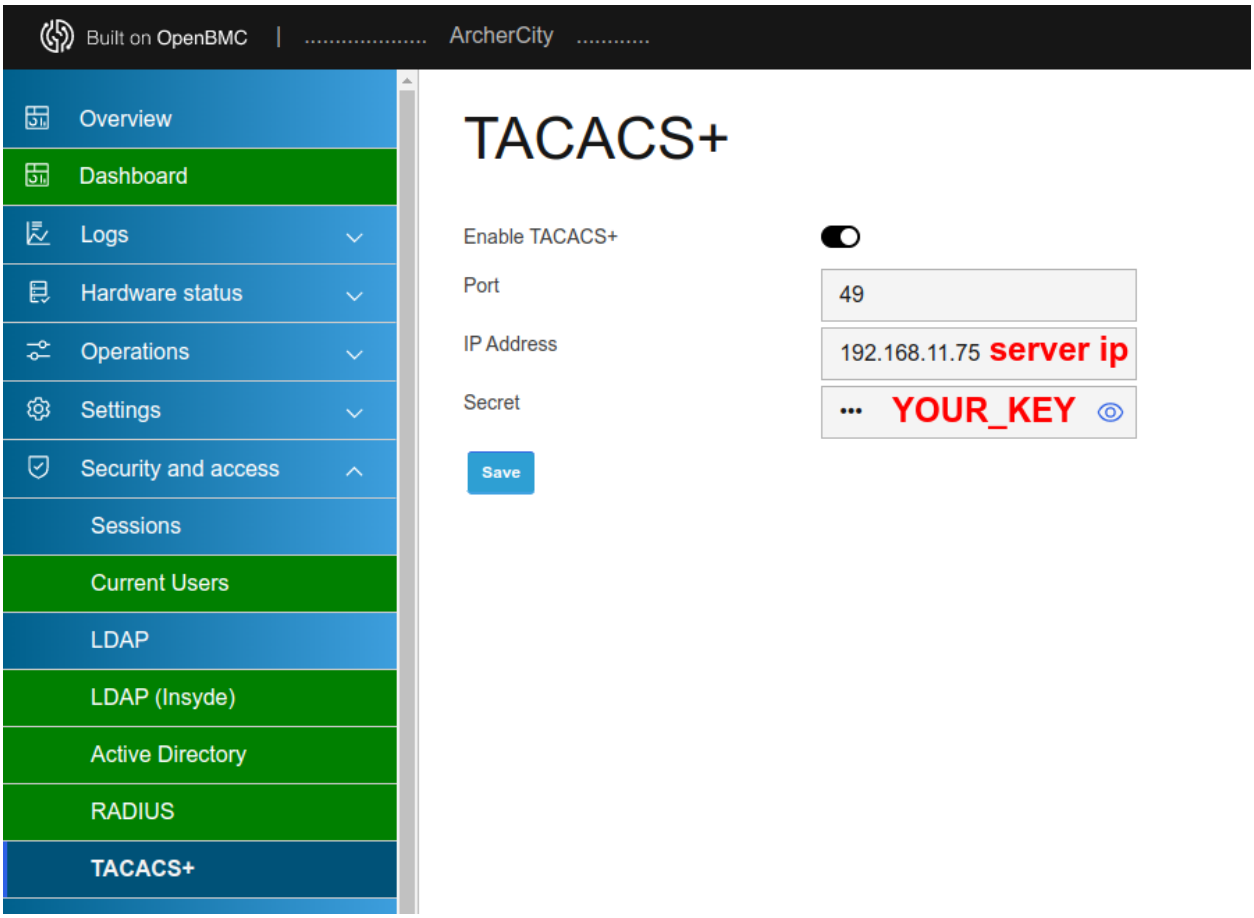
- Use the command **man tac_plus** to see the debug level

```
# sudo service tacacs_plus stop
# sudo tac_plus -C /etc/tacacs+/tac_plus.conf -d
```

3.4. TACACS+ authentication setting in BMC Web Service page

You will need to update the page in this picture

- **IPv4 Address**: ip of your TACACS+ server
- **Secret**: "YOUR_KEY" or your secret configured in the tac_plus.conf



- You can now login to BMC Web Service through the TACACS+ account
    - login with username **insyde_1** and password **PASSWD_1**
    - Note: Default port is 49

3.5. TACACS+ authentication setting by IPMI Commands

Setting TACACS+ in the Supervyse Web Service page will be completed,
for now, here is IPMI Commands: (Modify BMC_IP **192.168.11.209** and server ip **192 168 11 75** for your environment)

```
BMC_IP=192.168.11.209
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x67 1 1
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x67 3 192 168 11 75
ipmitool -I lanplus -C 17 -H $BMC_IP -U root -P 0penBmc raw 0x3e 0xc0 0xc 0x67 4 `python3 -c "print(' '.join(hex(ord(c)) for c in 'YOUR_KEY'.ljust(64, '\0')))"`
```

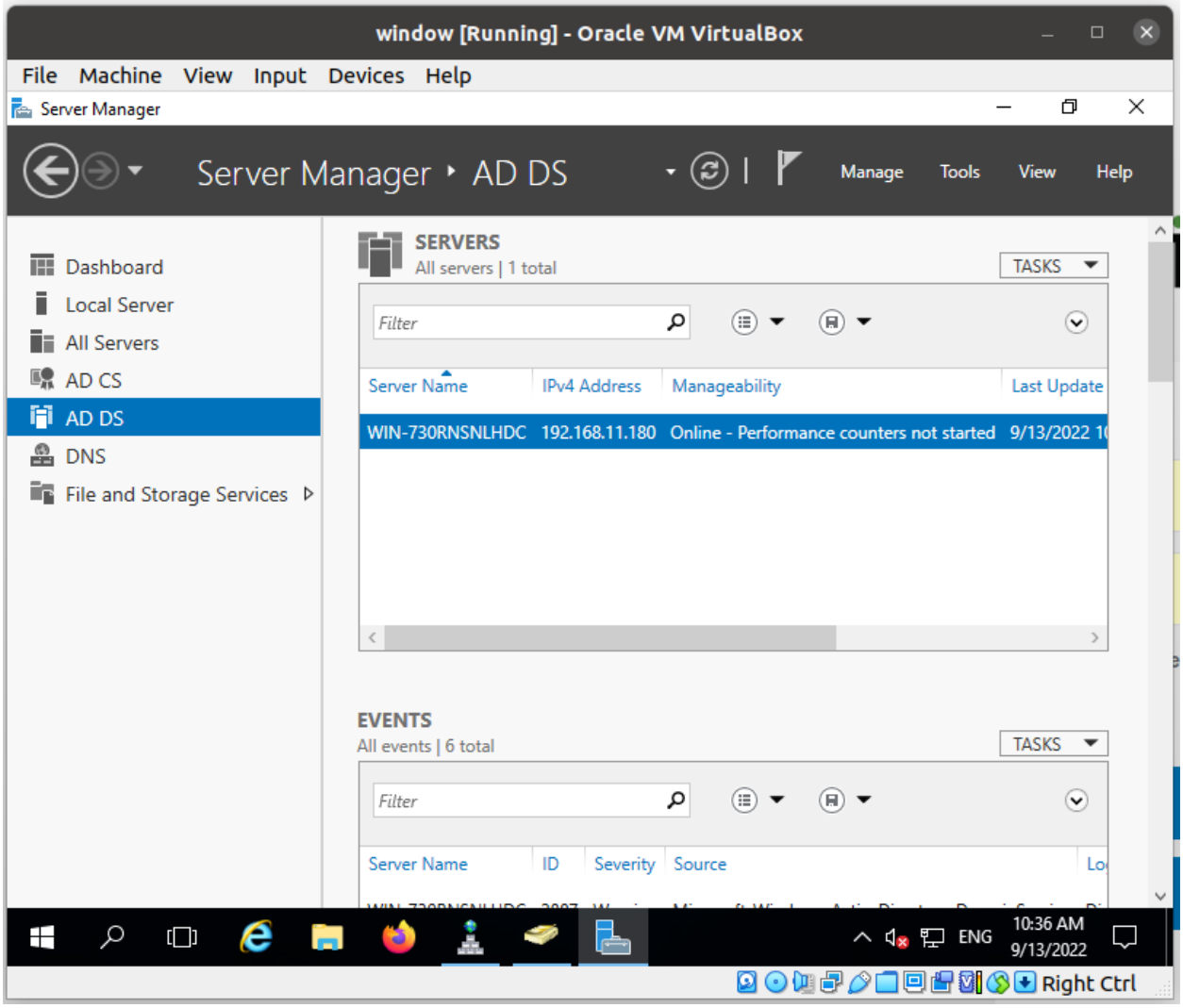
4. Supervyse OPF Active Directory Server Setting Guide

4.1. Introduction

- This guide introduces how to set up OPF Active Directory with Windows server 2016.
- Active Directory uses Active Directory Domain Services (AD DS) to authenticate user.
- Active Directory Domain Service(AD DS) is the foundation of every Windows domain network. It stores information about domain members, including devices and users, verifies their credentials, and defines their access rights.
- Set up Active Directory Domain Service(AD DS):
  [Active Directory Domain Services In Windows Server 2016]

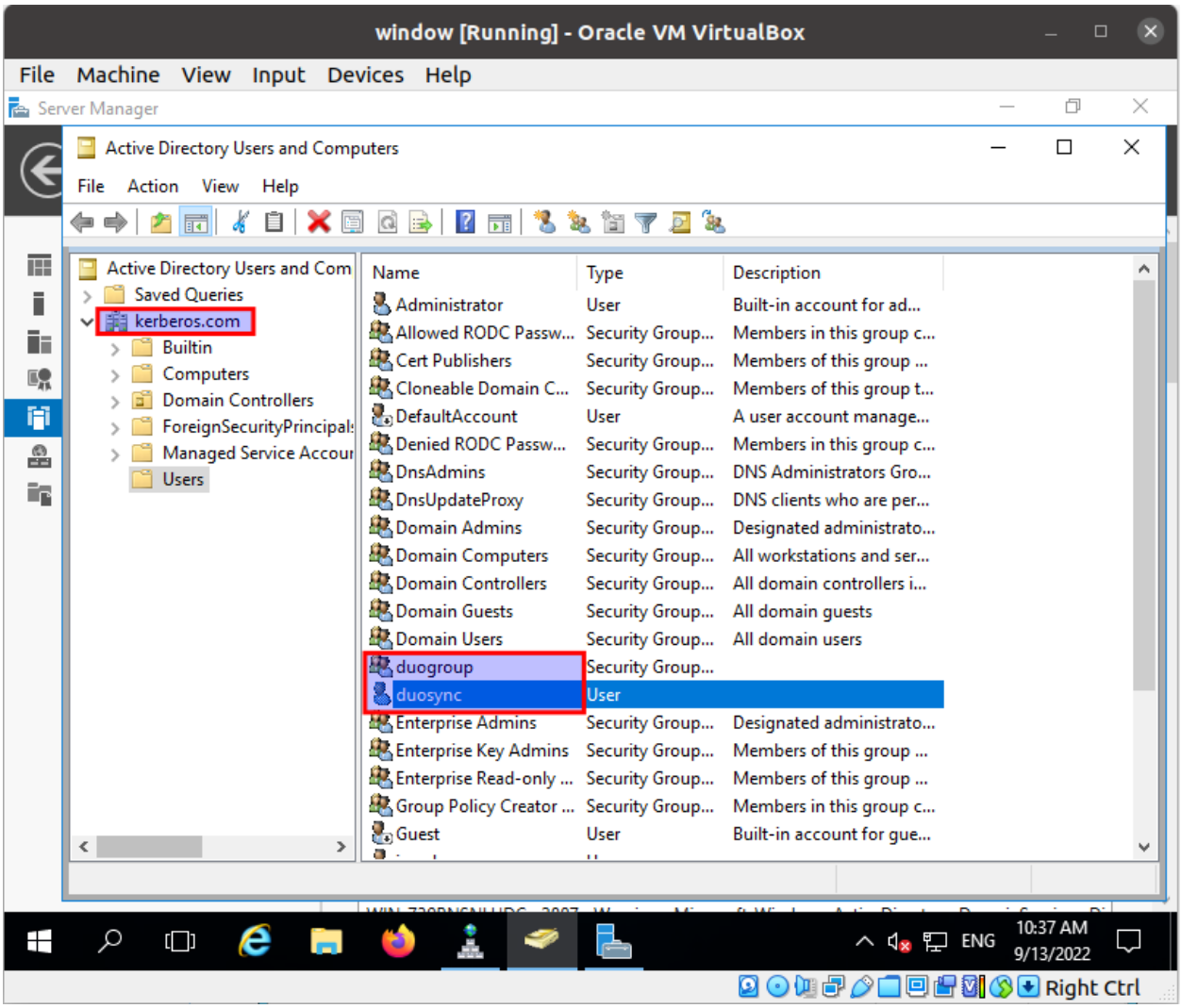
4.2. Configuring AD service

- Configure Active Directory server and create a domain **kerberos.com**

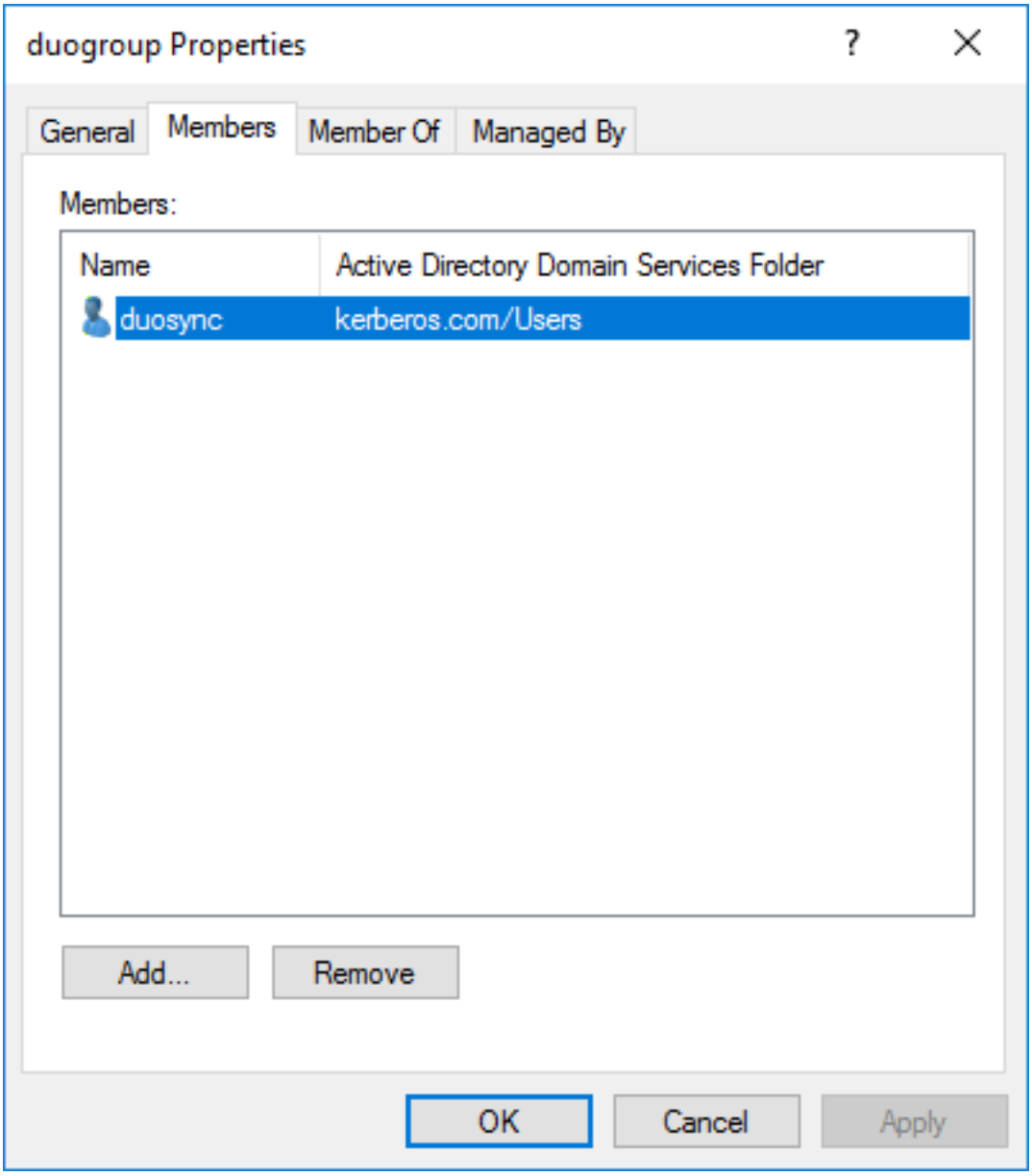


4.3. Configuring User and Group

- Open "Active Directory Users and Computers" in Server manager -> Tools
- Add group and user under domain kerberos.com
    - Add new group "duogroup"
    - Add new user "duosync"



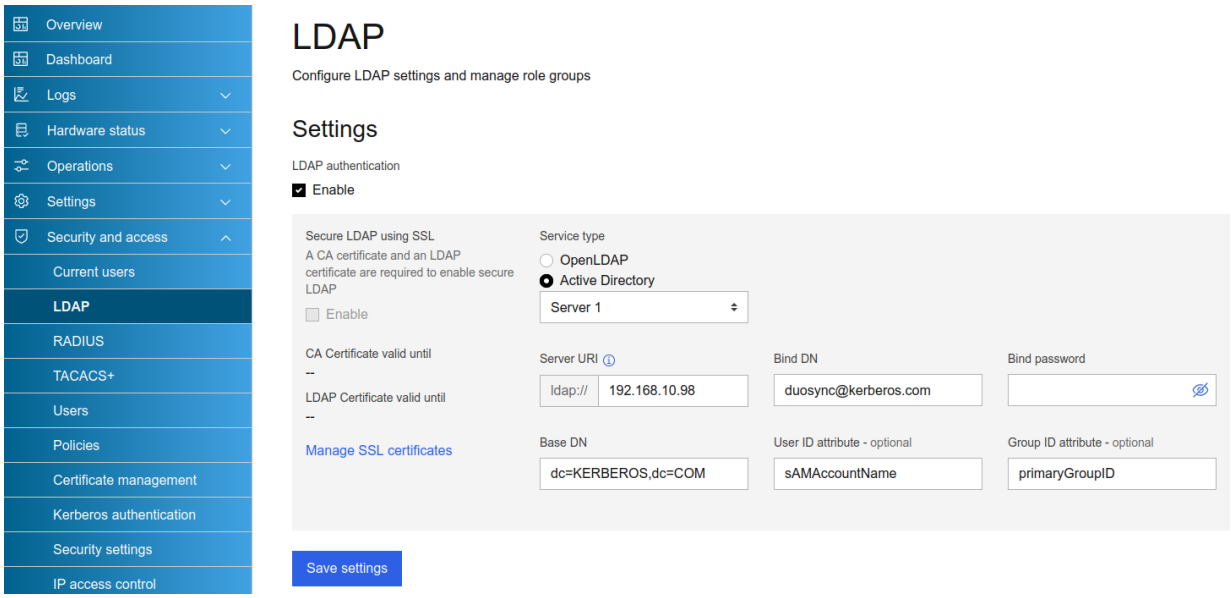
- Add user "duosync" as the member of "duogroup" group

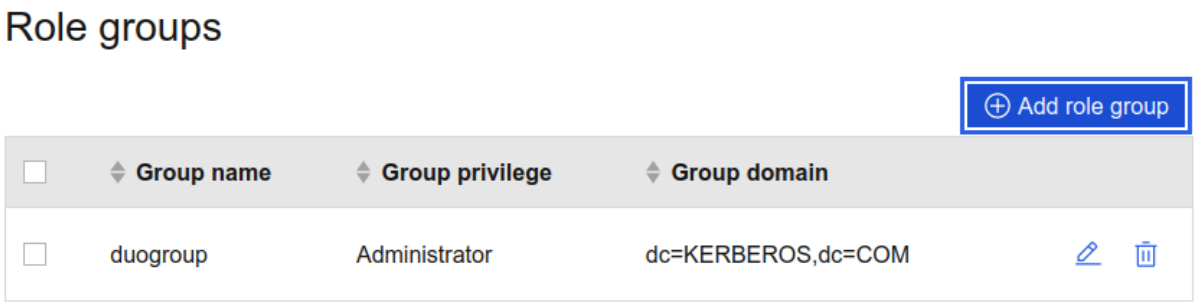


4.4. Configuring Bmcweb Active Directory

- Bmcweb security-and-access/ldap page
- Settings
    - Service type: Active Directory
    - Server URI: Your AD server IP
    - Bind DN: (AD user name)@(AD domain name) e.g. duosync@KERBEROS.COM
    - Bind password: AD user password
    - Base DN: AD domain name e.g. dc=KERBEROS,dc=COM
    - User ID and Group ID is optional and could keep empty

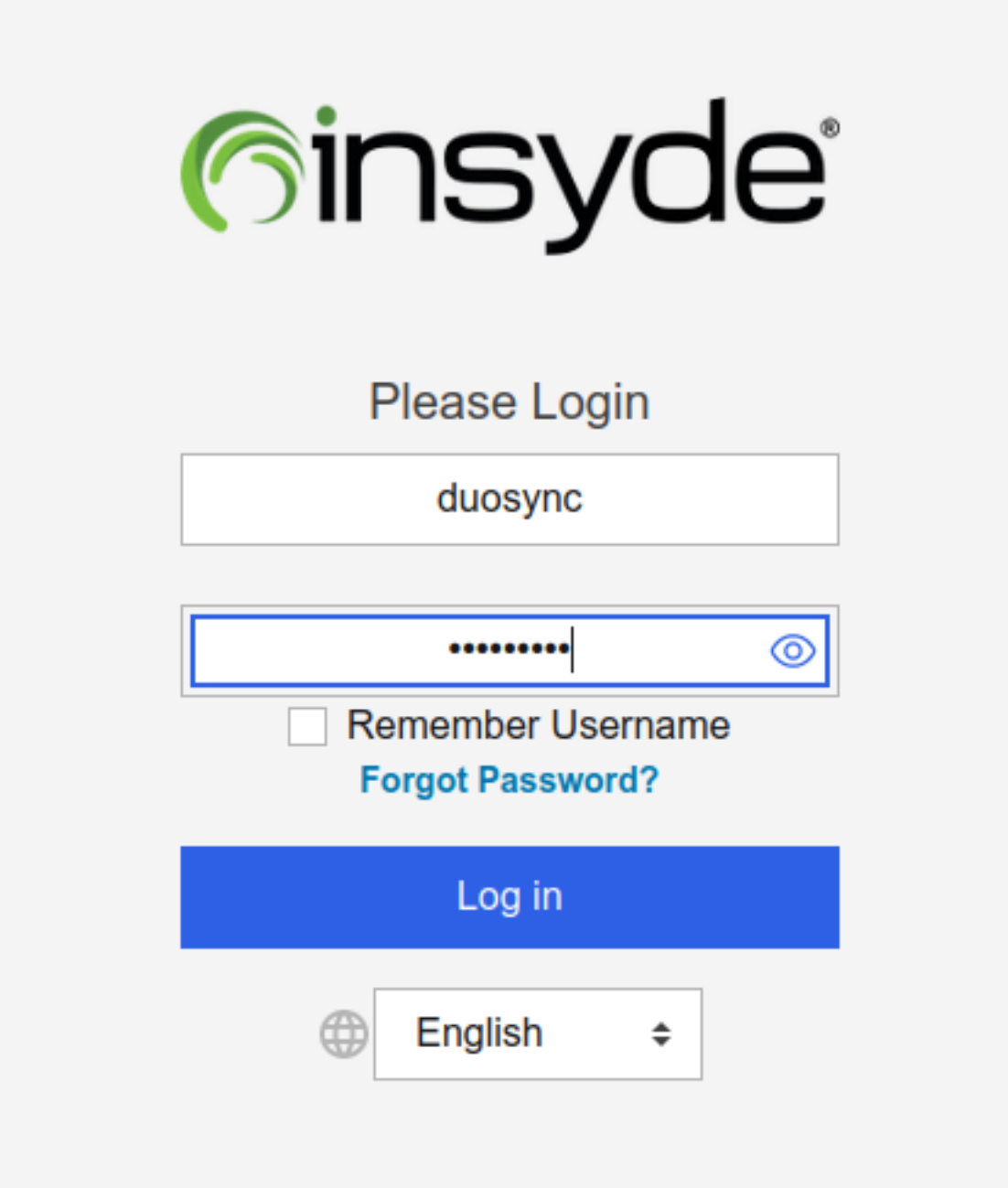
- Role groups
    - Add role group
    - Group name: Your AD user primary group e.g. duogroup
        - If you didn't setup AD user's primary group, default primary group will be **Domain Users**
    - Group privilege: Select login access ability e.g. Administrator
    - Group domain: Your AD group domain, LDAP distinguished name format e.g. dc=KERBEROS,dc=COM



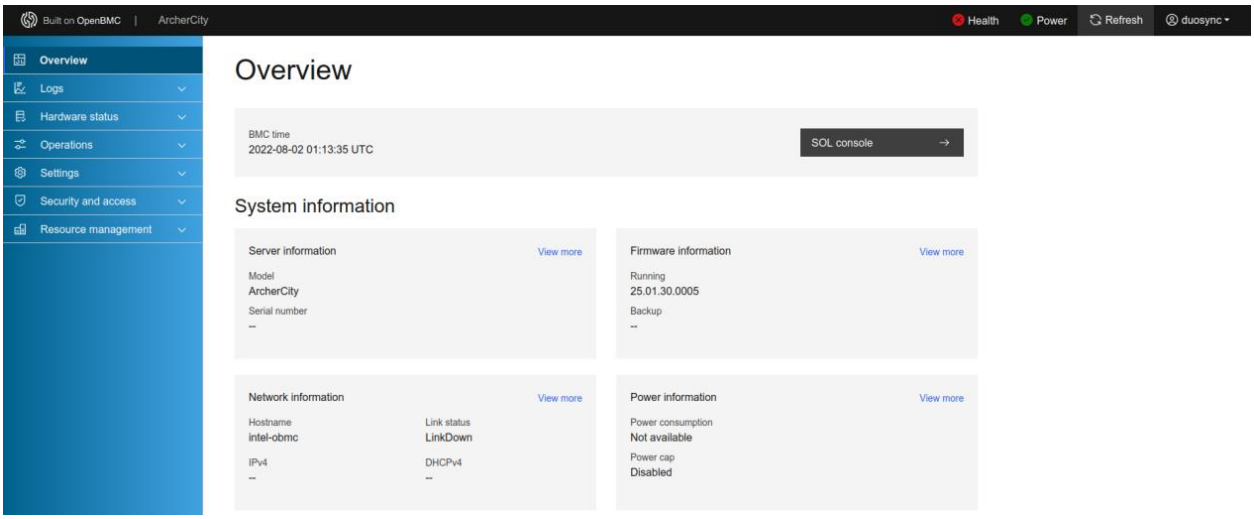


4.5. Active Directory Login

- Bmcweb login with AD user and password



- Bmcweb login with AD user success
    - The login user name should be your AD user name, e.g. duosync


Built on OpenBMC | ArcherCity
Health
Power
Refresh
duosync
Overview
Logs
Hardware status
Operations
Settings
Security and access
Resource management
Overview
BMC time
2022-08-02 01:13:35 UTC
SOL console
System information
Server information
View more
Model
ArcherCity
Serial number
--
Firmware information
View more
Running
25.01.30.0005
Backup
--
Network information
View more
Hostname
intel-obmc
Link status
LinkDown
IPv4
--
DHCPv4
--
Power information
View more
Power consumption
Not available
Power cap
Disabled

5. Supervyse OPF LDAP Server Setting Guide

5.1. LDAP

Lightweight **D**irectory **A**ccess **P**rotocol, the LDAP is an open, vendor-neutral, industry standard application protocol for accessing and maintaining distributed directory information services over an Internet Protocol network.

Directory services play an important role in developing intranet and Internet applications by allowing the sharing of information about users, systems, networks, services, and applications throughout the network.

In BMC, we use it to manage login accounts.

- [OpenLDAP]
  is an open source implementation of the Lightweight Directory Access Protocol.

LDIF fields

- dn: distinguished name
  This refers to the name that uniquely identifies an entry in the directory.

- dc: domain component
  This refers to each component of the domain. For example www.google.com would be written as DC=www,DC=google,DC=com

- ou: organizational unit
  This refers to the organizational unit (or sometimes the user group) that the user is part of. If the user is part of more than one group, you may specify as such, e.g., OU= Lawyer,OU= Judge.

- cn: common name
  This refers to the individual object (person's name; meeting room; recipe name; job title; etc.) for whom/which you are querying.

- sn: surname
  Specifies the person's surname, or last name. For example, sn: Anderson. A surname is required.

5.2. LDAP server on ubuntu

5.2.1. Environment setup

- OS : Ubuntu 20.04
- OpenLDAP : 2.4.49
- slapd : 2.4.49
- gnutls-bin : 3.6.13

- ssl-cert : 1.0.39

```
# sudo apt-get install slapd ldap-utils gnutls-bin ssl-cert
```

Prompt when installing:

6. Administrator password => **admin** (Please input your LDAP admin password)
7. Confirm password => **admin**

5.2.2. Configuring slapd

In Ubuntu, there is a terminal UI to reconfigure slapd,

```
# sudo dpkg-reconfigure slapd
```

Follow the steps below to set your environment:

8. Omit OpenLDAP server configuration? => **No**
9. DNS domain name => **sc1.test** (Domain name will be your base DN)
10. Organization name => **bmc**
11. Administrator password => **admin** (Please input your LDAP admin password)
12. Confirm password => **admin**
13. Do you want the database to be removed when slapd is purged => **No**
14. Move old database => **yes**

5.2.3. LDAP Browser/Editor

After configuring LDAP is finished, reviewing or editing the LDAP tree can be done through the LDAP browser/editor application. We will use **phpldapadmin** for the following example:

1. Install **phpldapadmin** for LDAP management

```
# sudo apt-get install phpldapadmin
```

2. Edit phpldapadmin configurations

```
# sudo vi /etc/phpldapadmin/config.php
```

```
...
/* Array of base DNs of your LDAP server. Leave this blank to have phpLDAPadmin
  auto-detect it for you. */
$servers->setValue('server','base',array('dc=sc1,dc=test'));

...

/* The DN of the user for phpLDAPadmin to bind with. For anonymous binds or
  'cookie','session' or 'sasl' auth_types, LEAVE THE LOGIN_DN AND LOGIN_PASS
  BLANK. If you specify a login_attr in conjunction with a cookie or session
  auth_type, then you can also specify the bind_id/bind_pass here for searching
  the directory for users (ie, if your LDAP server does not allow anonymous
  binds. */
$servers->setValue('login','bind_id','cn=admin,dc=sc1,dc=test');

...

/* Hide the warnings for invalid objectClasses/attributes in templates. */
```

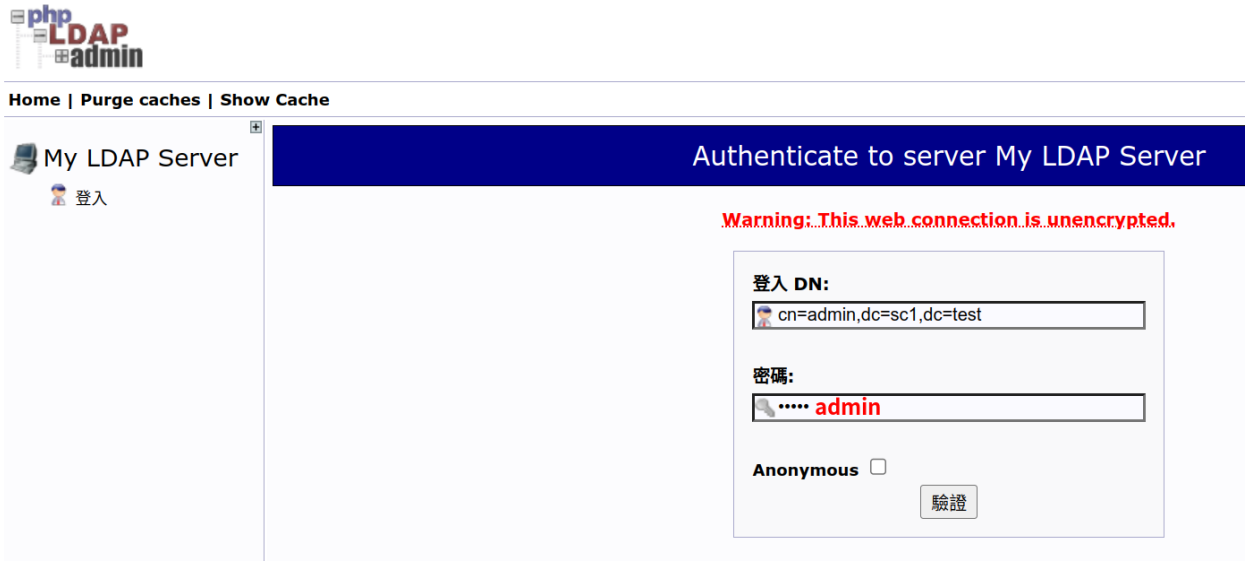
```
$config->custom->appearance['hide_template_warning'] = true;

...
```

3. Login to phpLDAPadmin to set LDAP users

- http://localhost/phpldapadmin

4. Login with password **admin**



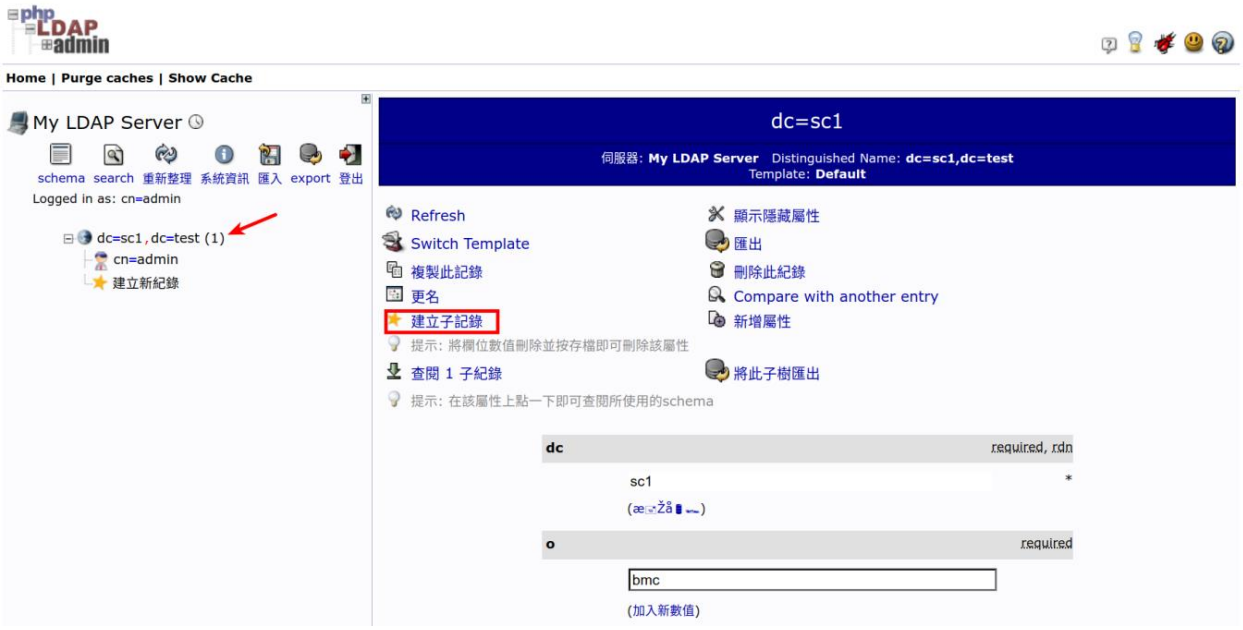
5.2.4. Example for add user and set privilege group

Below is example of add LDAP user via phpLDAPadmin

- add user **usr1**, password **pwd1** with **administrator** privilege.

5.2.4.1. Setting on LDAP server

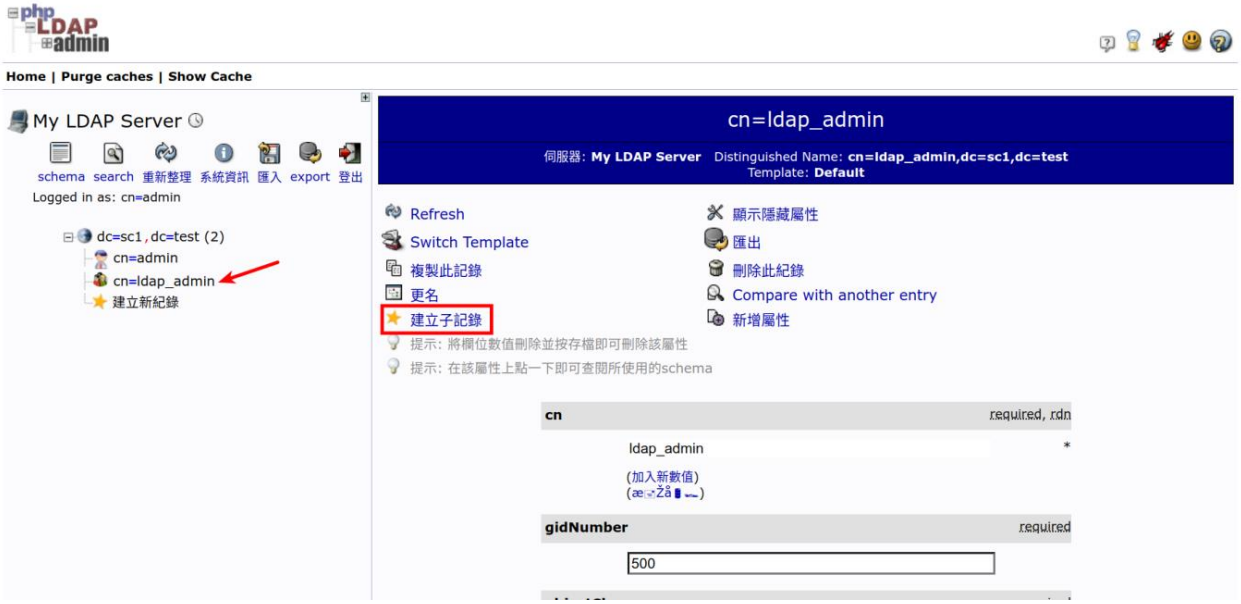
1.Unfold **dc=sc1,dc=test** folder and click **Create a child entry**



3.Input **ldap_admin** and click Create object

Note: This object will be group when BMC decide the privilege.

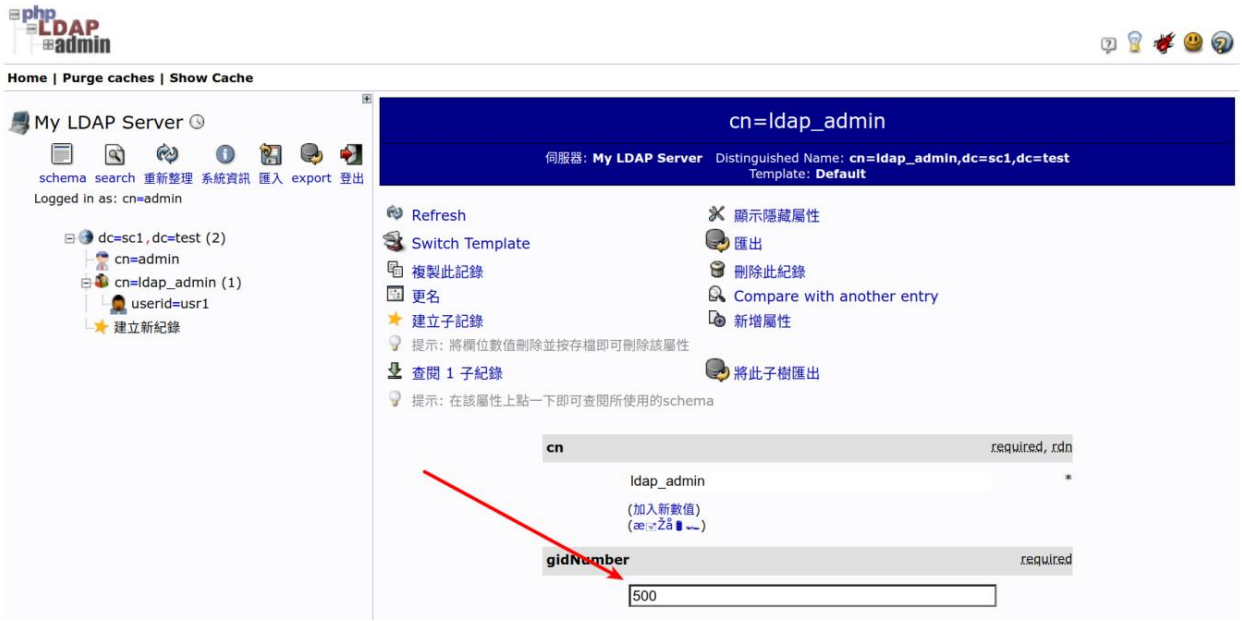
4.click cn=ldap_admin and click Create a child entry



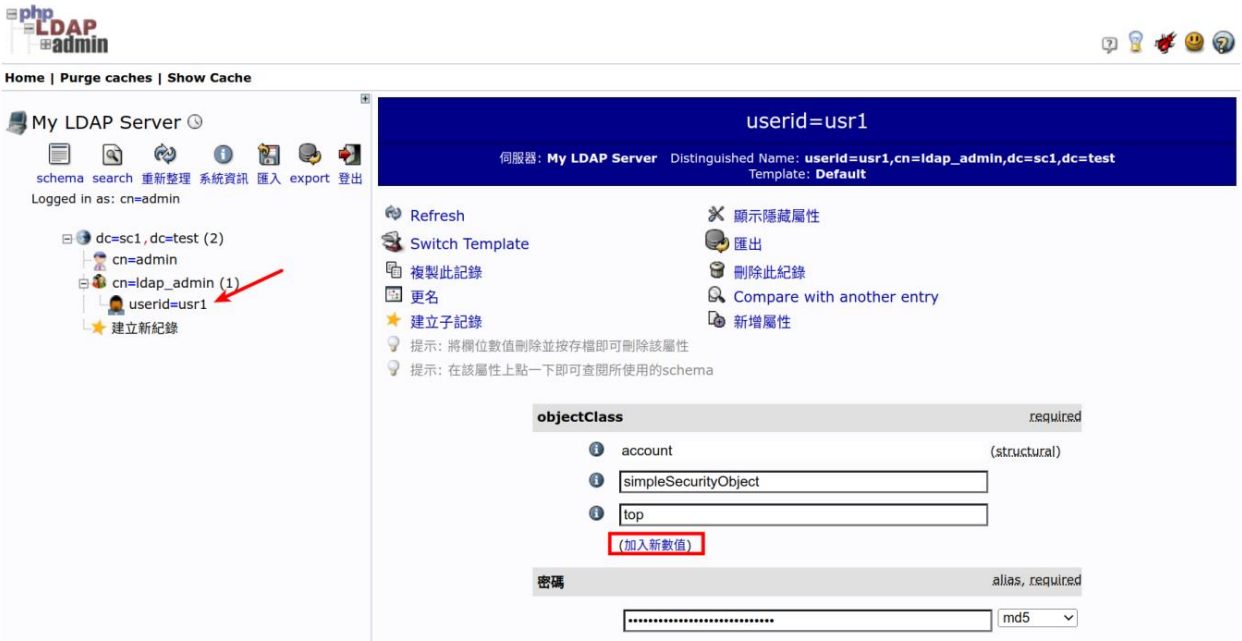
5.Click **Generic: Simple Security Object**

6.Input password **pwd1** and User Name **usr1**

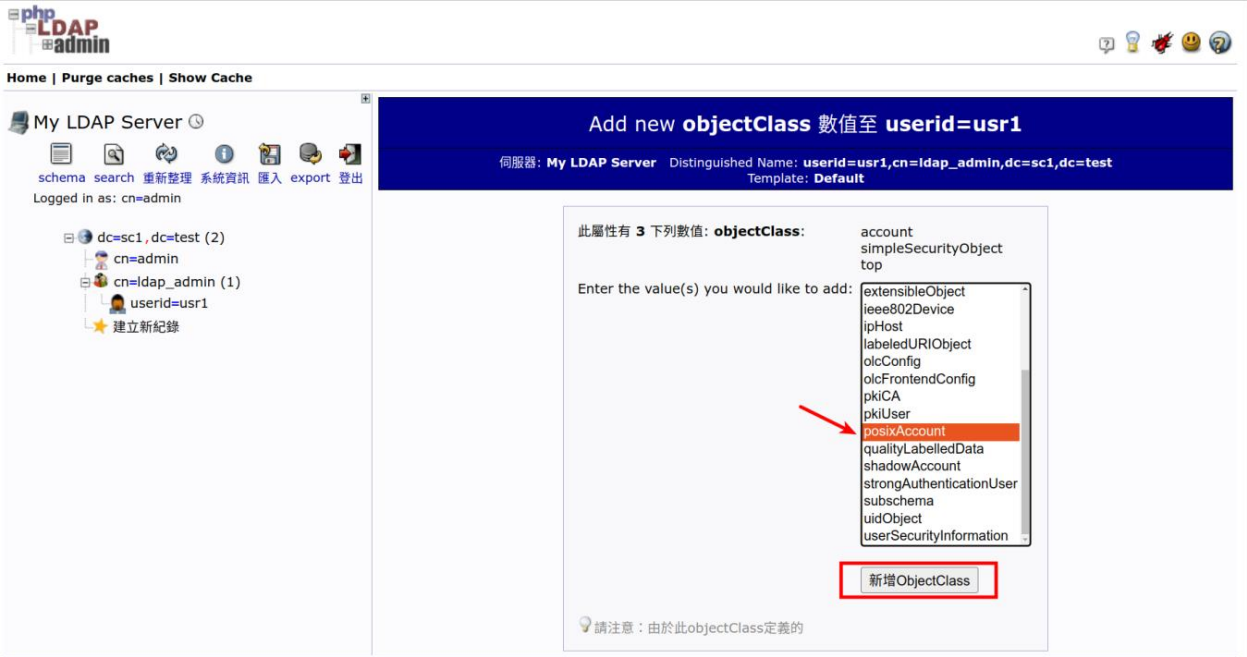
7.click cn=ldap_admin and check the **gid**, it's **500** in this example



8.click userid=usr1 and add new value to attribute objectClass

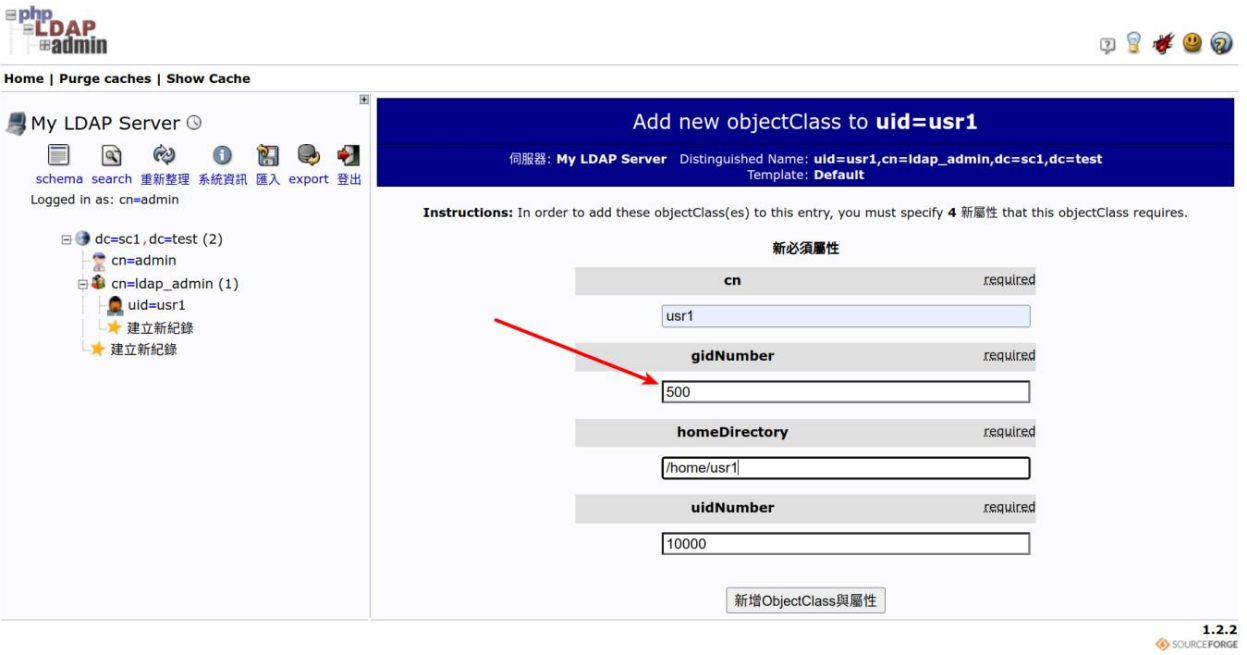


9.select **posixAccount**



10.Input attribute

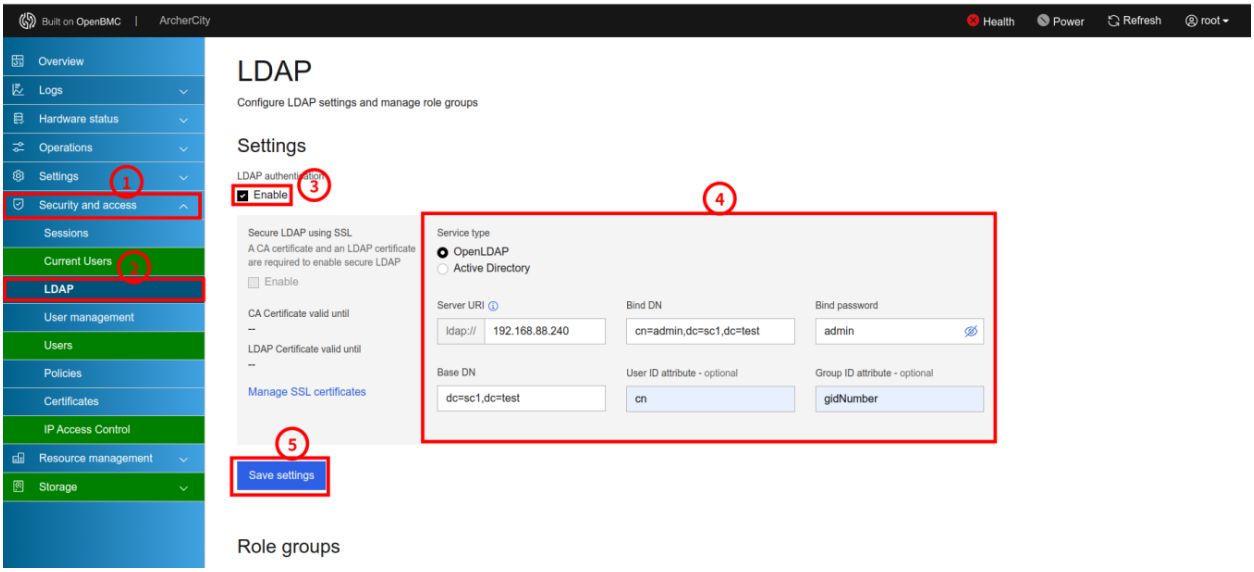
- cn is **usr1**
- gidNumber is the gid of ldap_admin, it's **500** in this example
- homeDirectory is **/home/usr1** (no effect)
- uidNumber is **10000** (no effect)



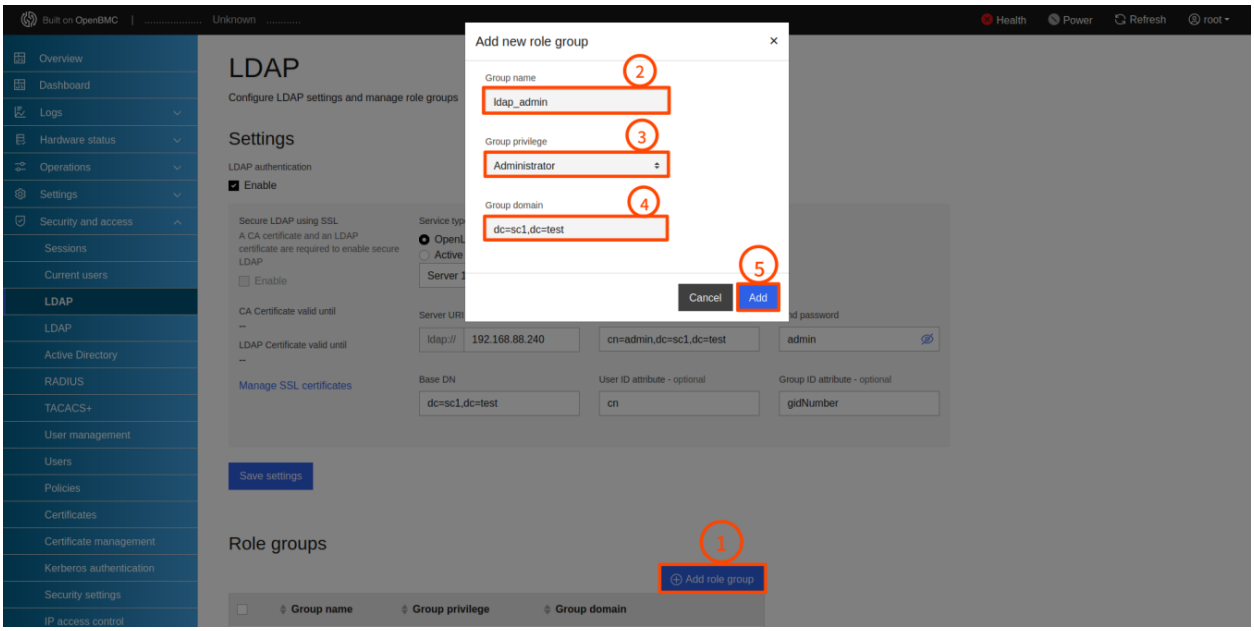
11 Now, LDAP server has settings done!

5.2.4.2. Setting on OPF BMC

- Web login -> Security and access Page -> LDAP page
- Enable LDAP authentication
- Choose Service type **OpenLDAP**
- Enter LDAP Server URI
- Bind DN is **cn=admin,dc=sc1,dc=test**
- Bind Password is **admin**
- Bind DN is **dc=sc1,dc=test**
- User ID attribute is **cn**
- Group ID attribute is **gidNumber**
- click Save settings, then you can add role group



- click Add role group
- Group name is **ldap_admin**
- Group privilege **Administrator**
- Group domain is **dc=sc1,dc=test**
- click Add



5.2.4.3. Login Web test

- Now you can login with LDAP user **usr1** with password **pwd1**

6. Supervyse OPF Kerberos SSO Login Setting Guide

6.1. Kerberos

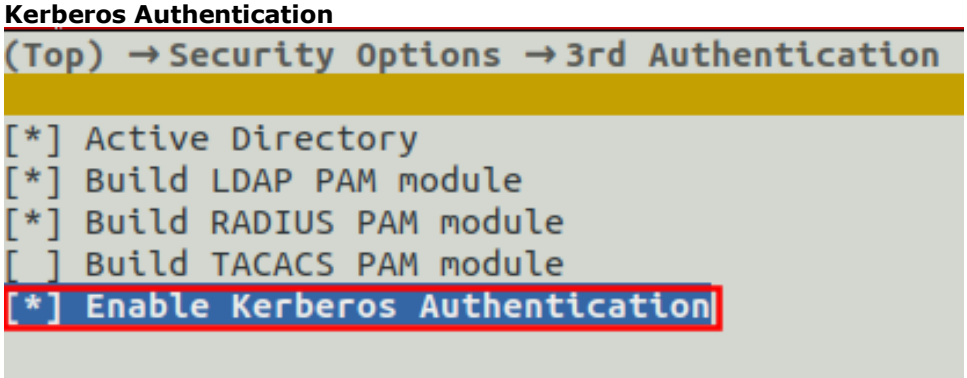
6.1.1. Introduction

- This page is the Kerberos SSO Login setting guide for Windows Server 2016, Ubuntu client and BMC.
- Kerberos uses Active Directory Certificate Services (AD CS) to authenticate user.
- Active Directory Certificate Services (AD CS) establishes an on-premises public key infrastructure. It can create, validate, revoke and perform other similar actions, public key certificates for internal uses of an organization.

6.1.2. Prerequisites

Kerberos Authentication is default enabled
If CONFIG_KRB5, CONFIG_SASL are disabled in menuconfig .config file, follow below steps to re-enable them.

1. Ensure Kerberos Authentication is existed on the BMC firmware image
   - OPF could use pkconfig to modify menuconfig, e.g. bitbake pkconfig -c menuconfig
     - **menuconfig -> Security Options -> 3rd Authentication -> Enable Kerberos Authentication**



- **menuconfig -> Network Services -> Web Server -> WEB STYLE ->**

```
(Top) →Network Services →Web Server →WEB STYLE →Config WEB Pages →Configur

[*] Alerts
[*] Alert Email
[*] Date & Time
    IPv4 Network  --->
    IPv6 Network  --->
[*] VLAN
[*] LDAP
[*] RADIUS
[*] KERBEROS AUTH
[*] Active Directory
[*] SSL Certification
[*] Users
[*] Security Settings
[*] SOL & SMASH
    Firmware Update  --->
[*] IP Access Control
[*] Dynamic DNS
[*] Boot
[*] KVM & Media
```

  - Choose the menuconfig option to enable the Kerberos authentication for the generated BMC firmware image.

6.1.3. Windows server 2016

- **Setup AD DS and DNS**
  1. Open "Server Manager"
  2. Select "Add roles and features"
  3. Select "Active Directory Domain Services" and "DNS Server"
  4. Keep choosing "Next" then "Install"
  5. After Install, Click the "Flag" on top right of "Server Manager"
  6. Open the "Domain Controller" Setup window
  7. Create new Forest
  8. Name the domain e.g. "kerberos.com"
  9. Choose next to Complete and let the system do the setup
  10. System will need to reboot and you will be login as KERBEROS/Administrator

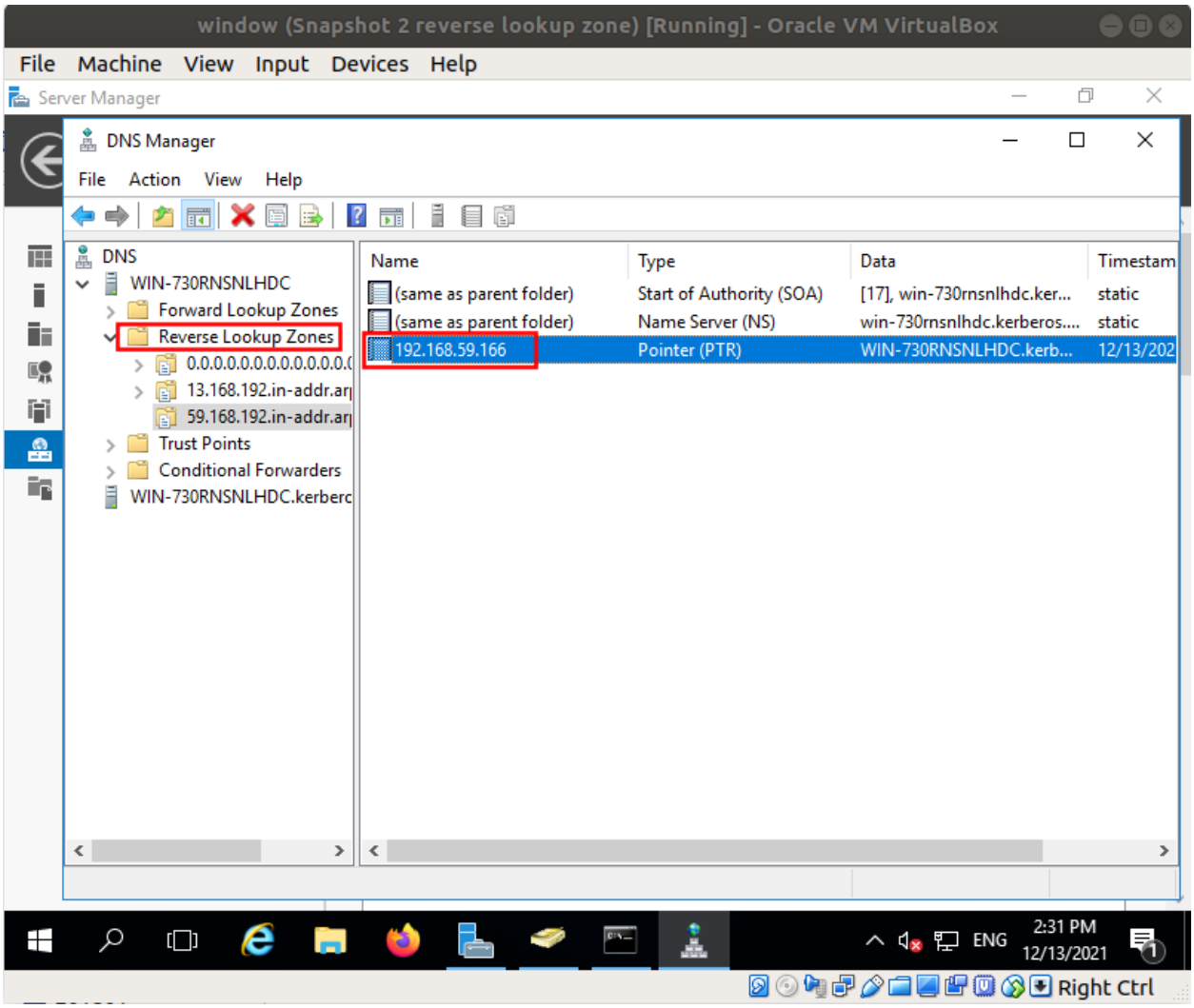
- **Setup AD CS**
  1. Open "Server Manager"
  2. Select "Add roles and features"

3. Select "Active Directory Certificate Services"
4. Select "Ceritication Authority"
5. Complete the installation
6. After Install, Click the "Flag" on top right of "Server Manager"
7. Open the "Certificate Authority" Setup window
8. Don't change anything to "Credentials" select next
9. Choose "Certificate Authority"
10. Choose "Enterprise CA"
11. Choose "Root CA"
12. Choose "Create a new private key"
13. Change algorithm to "SHA1"
14. Don't change anything on "CA Name"
15. Set "Validity Period" to 99 Years
16. Don't change anything on "Ceritifcate Database"
17. Review everything and select "Configure"
18. reboot system

[https://www.virtuallyboring.com/setup-microsoft-active-directory-certificate-services-ad-cs/]

- **Create required Active Directory users**
  AD DS -> Active Directory Users and Computers
  1. Create test AD user e.g. insyde
  2. Create user account for service(BMC)
     1. User account e.g. duosync
     2. This account will be linked with service principal name(SPN)
     - Could reference Active Directory Setup Guide **Configuring User and Group** option
- **Configure DNS**
  DNS -> DNS manager
  1. Forward lookup
     1. BMC hostname e.g. kdc01 -> BMC ip
  2. Reverse lookup
     1. windows server ip -> windows server hostname

In this example reverse lookup zone server ip will be your kdc server ip assigned by DHCP.



- **SetSPN**

Add service principal name to account
Open windows 2016 server cmdtool shell to use setspn command

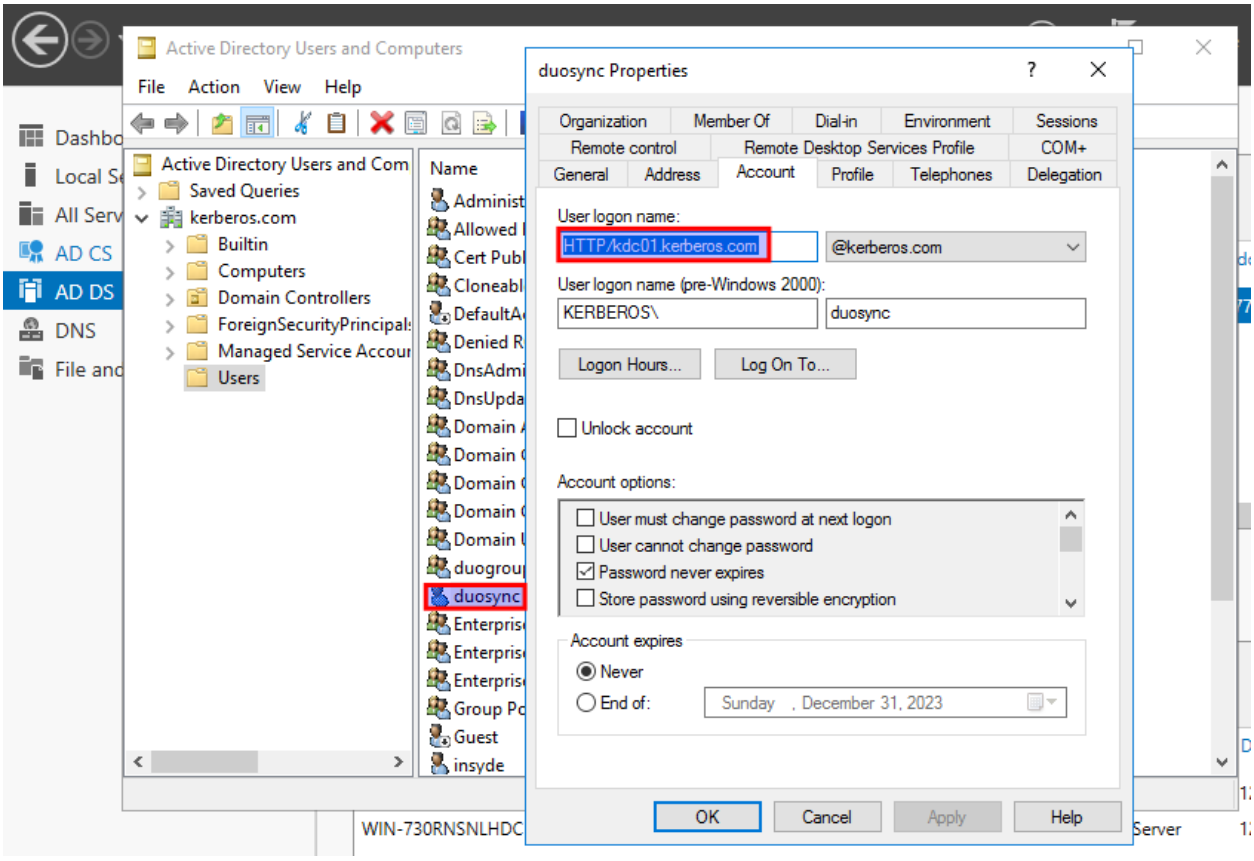
```
setspn -s SPN accountname
```

SPN - /@

```
setspn -s HTTP/kdc01.kerberos.com@KERBEROS.COM duosync
```

Windows 2016 server AD DS -> Active Directory Users and Computers -> Users -> account properties -> Account
Account duosync User logon name will become SPN.

e.g. HTTP/kdc01.kerberos.com



- **Create keytab file**

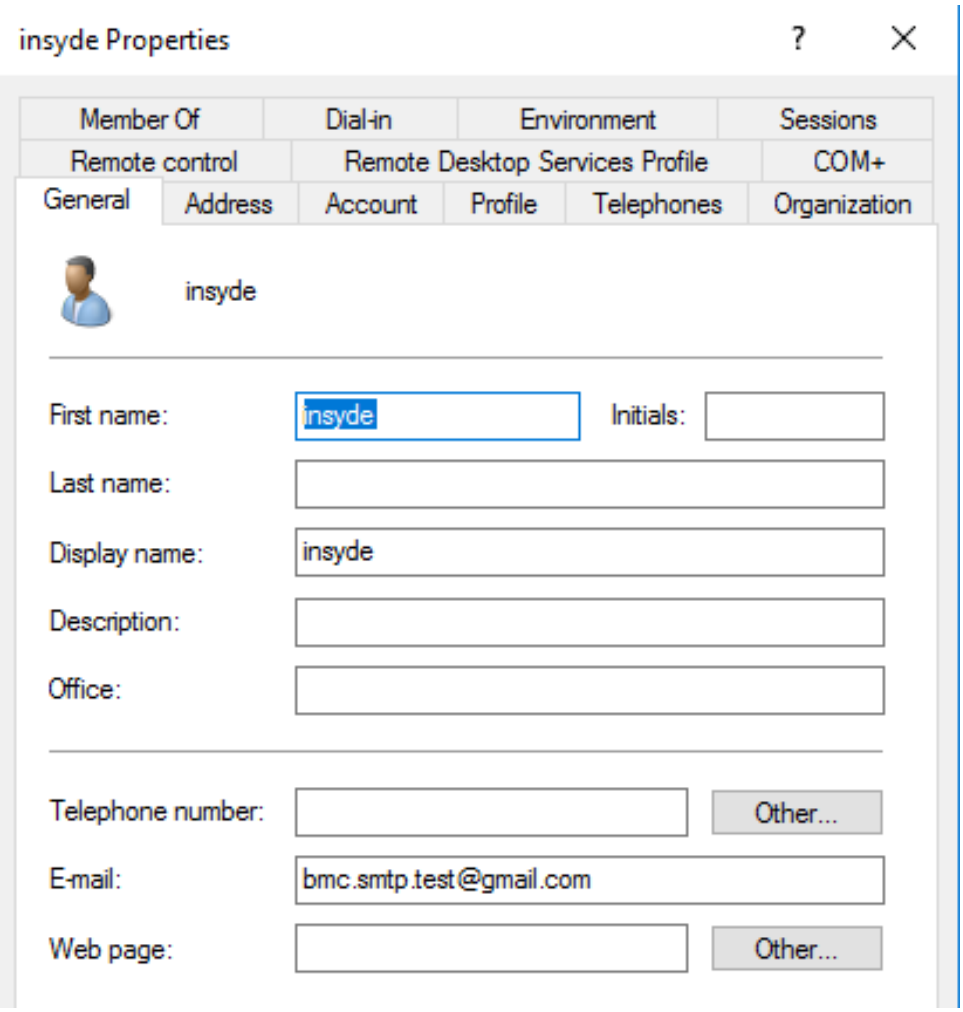
```
ktpass -princ <principalname> -mapuser <useraccount> -mapOp {set} -crypto {AES256-SHA1} -ptype
{KRB5_NT_PRINCIPAL} -pass <password> -out <filename>
```

<principalname> - <service class>/<BMC FQDN>@<domain name>

```
C:\> ktpass -princ HTTP/kdc01.kerberos.com@KERBEROS.COM -mapuser KERBEROS\duosync -mapOp set -crypto
AES256-SHA1 -ptype KRB5_NT_PRINCIPAL -pass Insyde760 -out c:\krbkeytab
```

- **AD user with email**
  This AD user email will be used for sending 2FA passcode

  AD DS -> Active Directory Users and Computers -> AD user properties -> E-mail



6.1.4. Ubuntu

- **Install Kerberos client**

```
sudo apt-get install krb5-user
```

Enable default Kerberos realm e.g. KERBEROS.COM

- **Edit file/etc/krb5.conf**

kdc and admin_server will insert window KDC server name or IP address

```
[libdefaults]
   default_realm = KERBEROS.COM
[realms]
   KERBEROS.COM = {
      kdc = win-730rnsnlhdc.kerberos.com
      admin_server = win-730rnsnlhdc.kerberos.com
   }
[domain_realm]
   .kerberos.com = KERBEROS.COM
   kerberos.com = KERBEROS.COM
```

- **Edit file /etc/hosts**

```
<BMC ip> kdc01.kerberos.com
e.g. 192.168.9.163 kdc01.kerberos.com
<Windows server ip> (window DNS server name)
e.g. 192.168.11.181 win-730rnsnlhdc.kerberos.com
```

- **NTP sync**

Setup Ubuntu client time sync with NTP server according to BMC NTP server setup.
Or setup the same NTP server manually for Ubuntu client and BMC.

E.g. BMC NTP sync from DHCP server is time1.google.com then setup Ubuntu client also sync time from the same NTP server

If NTP server is not the same or there is sync issue, kerberos validate may fail

```
sudo ntpdate time1.google.com
```

- **Kinit and Klist**

kinit - Get Kerberos TGT from KDC server

```
kinit insyde
```

klist - list credentials

```
klist

Ticket cache: FILE:/tmp/krb5cc_1000
Default principal: insyde@KERBEROS.COM

Valid starting     Expires            Service principal
2020-03-27T15:17:18  2020-03-28T01:17:18  krbtgt/KERBEROS.COM@KERBEROS.COM
    renew until 2020-04-03T15:17:15
```

- **Browser Setup**
  - **Firefox**

  1. Search enter about:config

  2. Search network.negotiate-auth.trusted-uris
  3. Enter the BMC domain e.g. kdc01.kerberos.com

  - **Chrome**

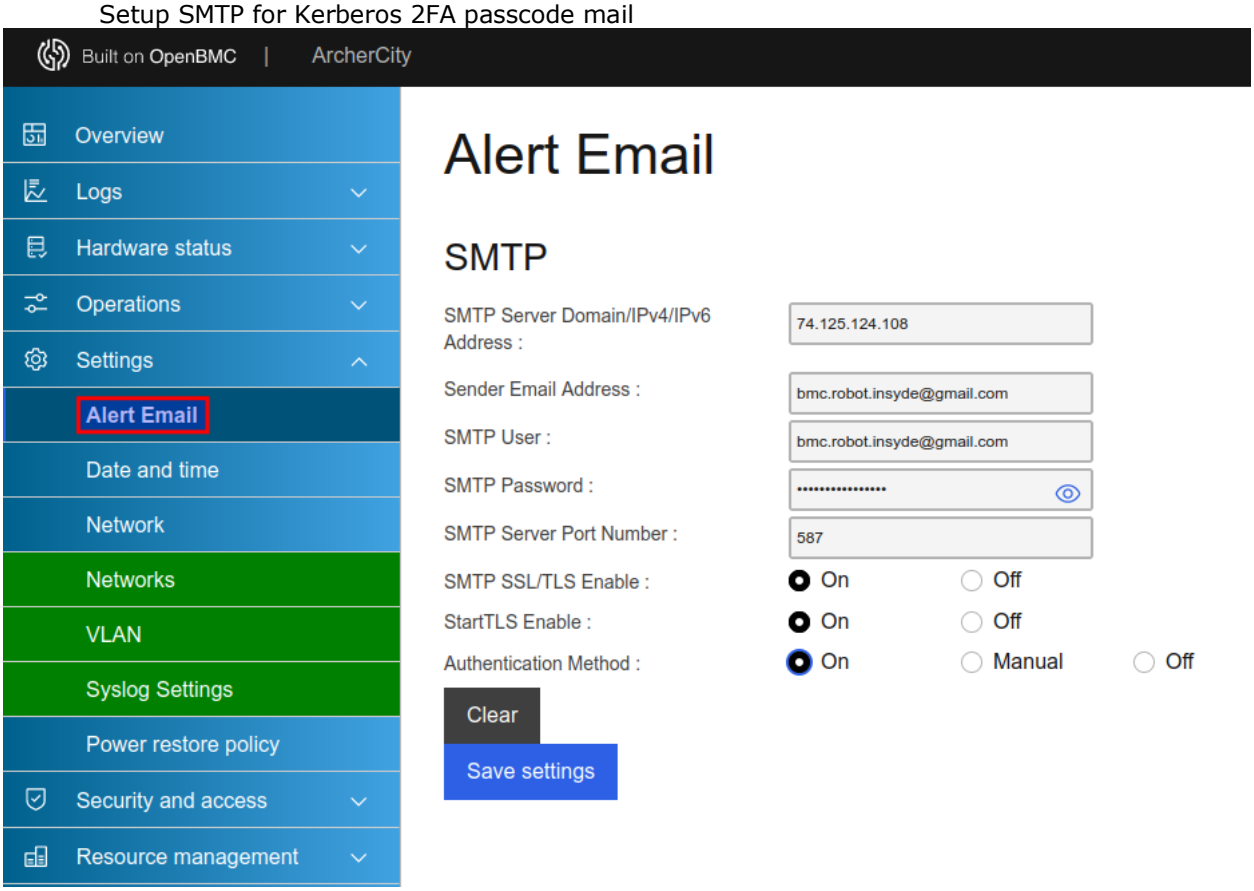
  1. Add json file under /etc/opt/chrome/policies/managed
     - E.g. kerberos_policy.json

```
{
  "AuthServerWhitelist" : "kdc01.kerberos.com",
  "AuthNegotiateDelegateWhitelist" : "kdc01.kerberos.com",
  "DisableAuthNegotiateCnameLookup" : true,
  "EnableAuthNegotiatePort" : true
}
```

6.1.5. BMC Setup

Use Firefox or Chrome to visit BMC Web by FQDN e.g. kdc01.kerberos.com

- **Settings -> Alert Email**
  Setup SMTP for Kerberos 2FA passcode mail



- **Settings -> Date and time**
  Setup the same NTP server like Ubuntu client
  Choose one of below setup cases for NTP server

  1. Network is DHCP

     1. USE NTP servers(DHCP) is enable

        - Automatically sync NTP server from DHCP server and no need to change

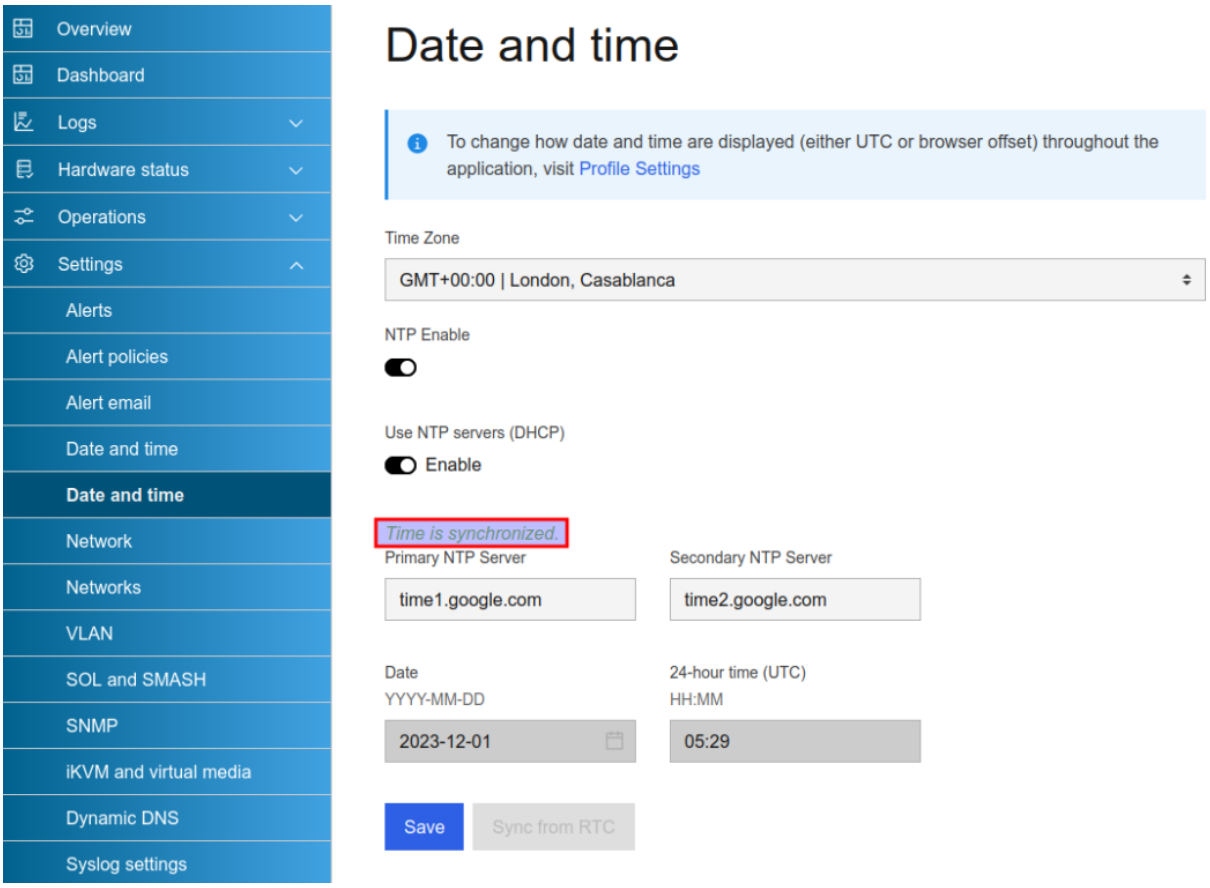
     2. USE NTP servers(DHCP) is disable

        - Manually setup NTP server

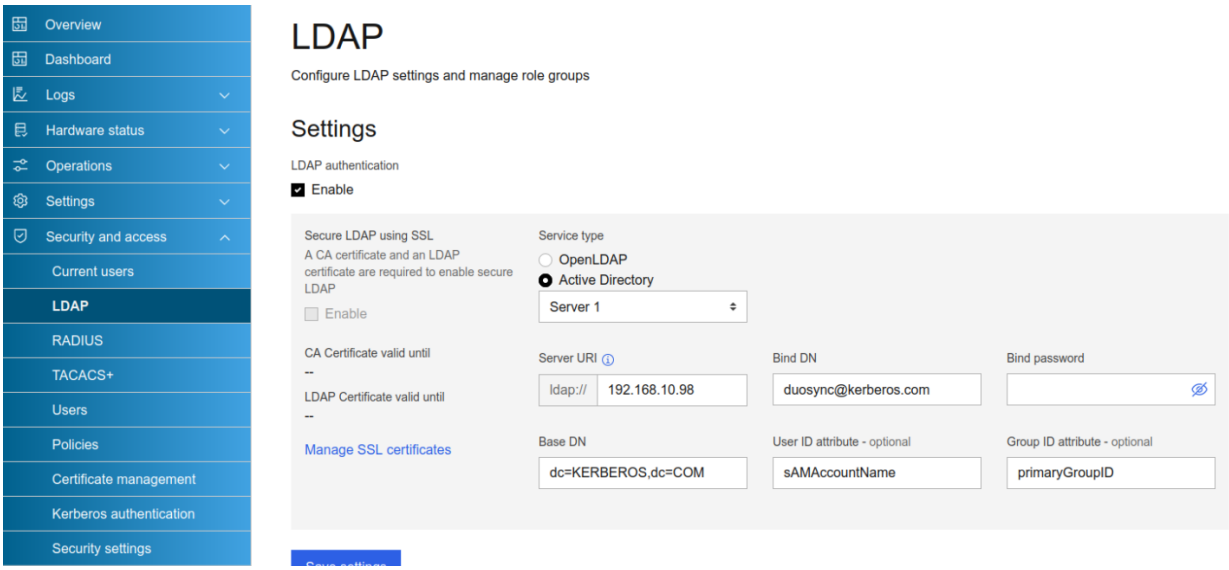
  2. Network is static and USE NTP servers(DHCP) is disable

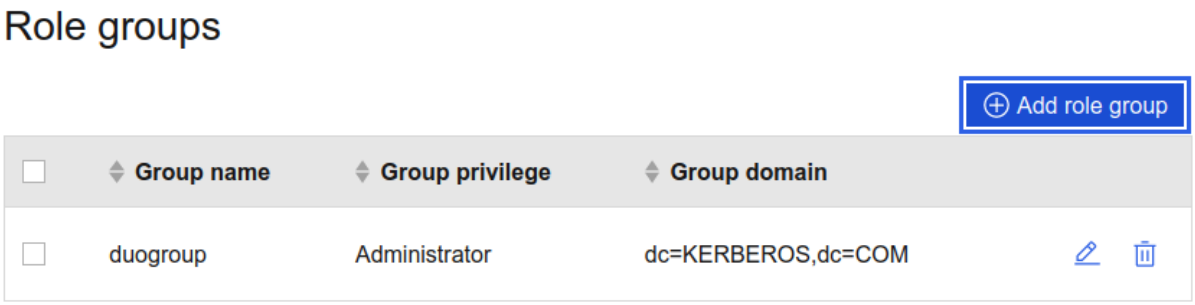
     - Manually setup NTP server

Make sure after any of above NTP server setup, time is synchronized



- **Security and access -> LDAP**
  1. Enable LDAP authentication
  2. Select Service type Active Directory
  3. Insert Server URI, Bind DN, Bind password, Base DN then save settings
  4. Click Add role group
     - Group name e.g. duogroup
     - Decide user login privilege e.g. Administrator

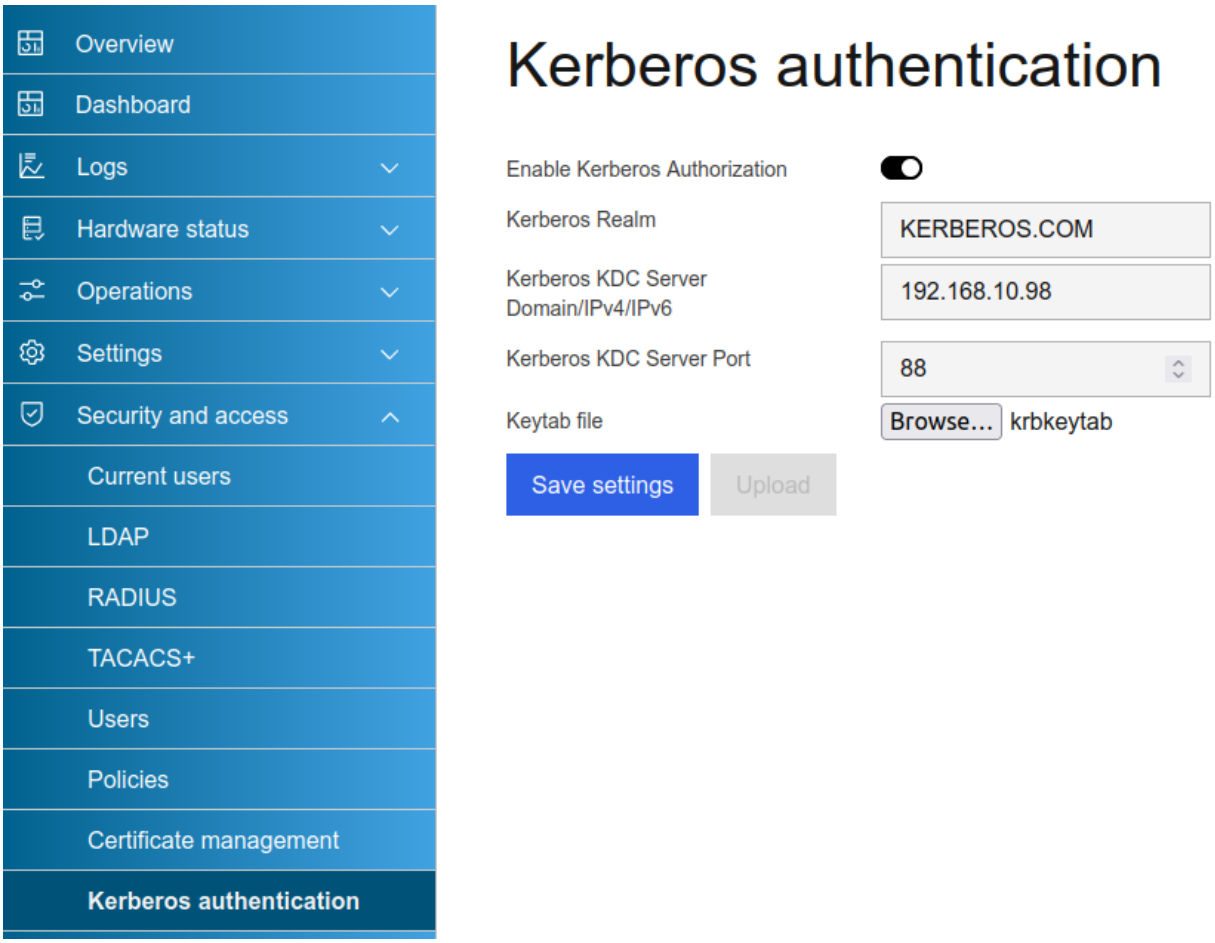




- **Security and access -> Kerberos Authentication**

  Setup realm, KDC server address, KDC server port.
  Save settings

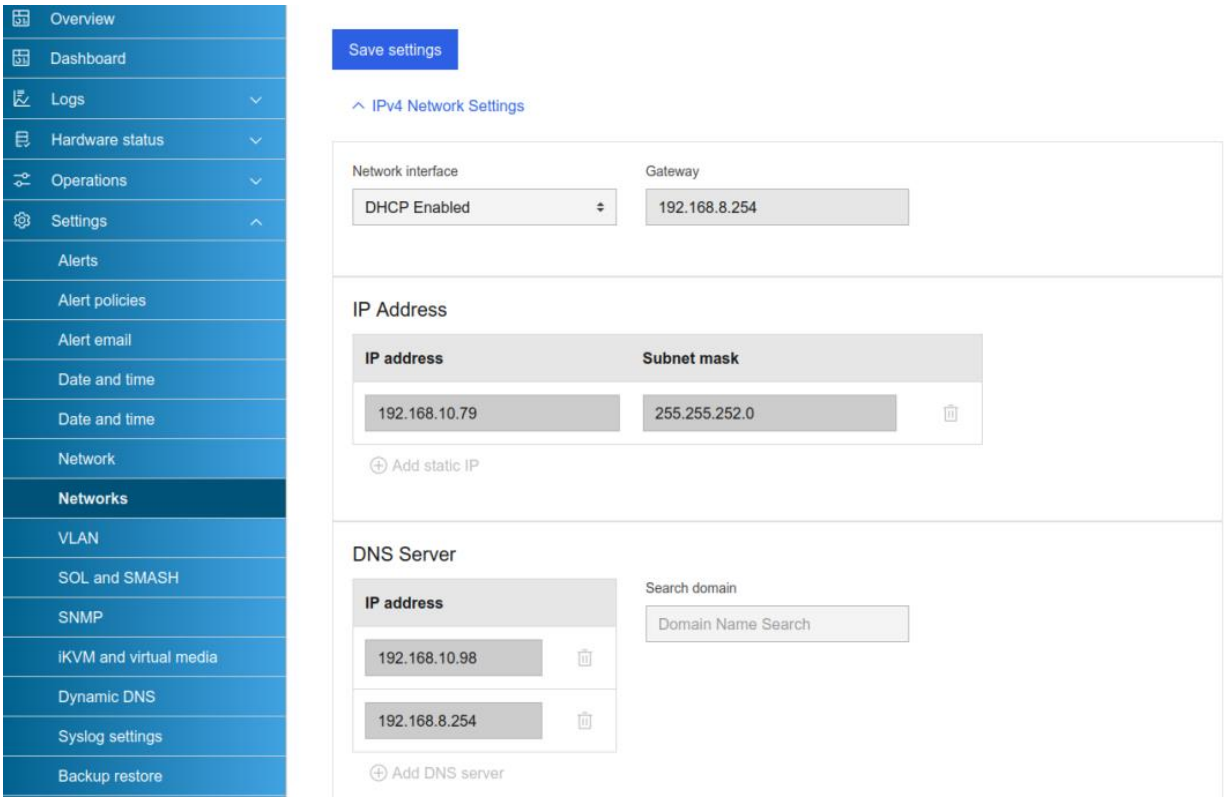
  Choose keytab File then Upload.



- **Settings -> Networks -> IPv4 Network Settings**

  Setup DNS server for window 2016 KDC server recognizing
  Choose Network interface DHCP or static

  1. DHCP
     1. Network interface -> Static IP Address
     2. DNS Server -> Window 2016 KDC server IP
     3. BMC may logout automatically then login to IPv4 Network Settings again
     4. Network interface -> DHCP

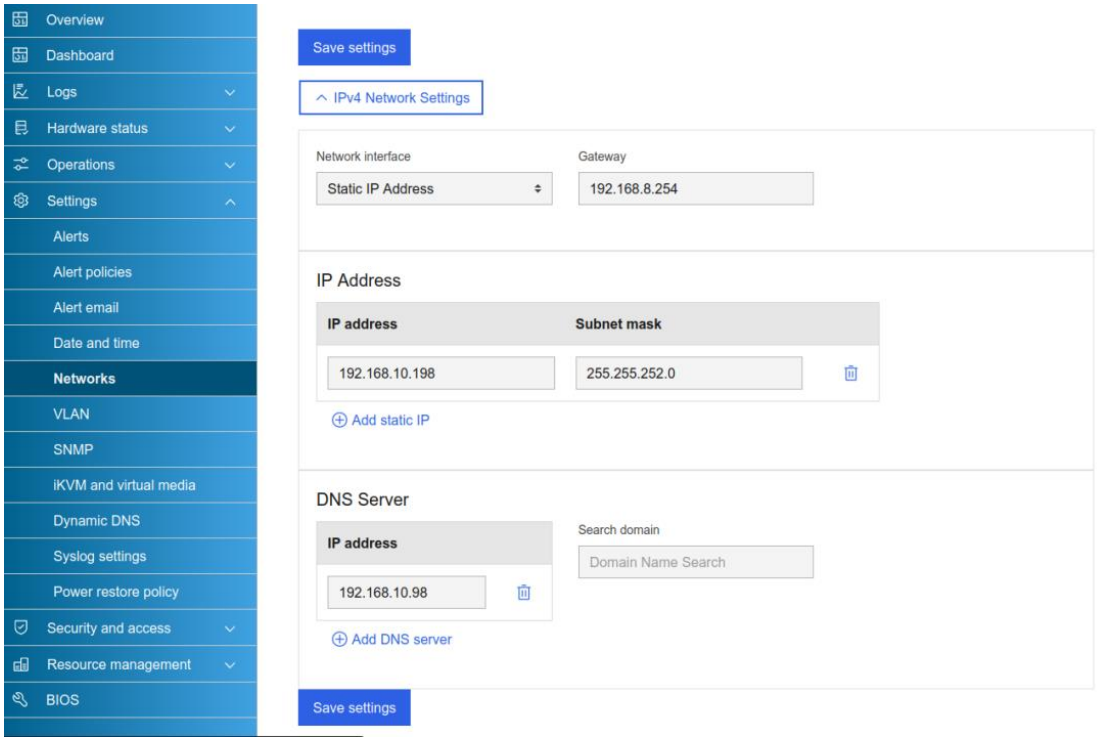


2. Static

    1. Network interface -> Static IP Address

    2. DNS Server -> Window 2016 KDC server IP

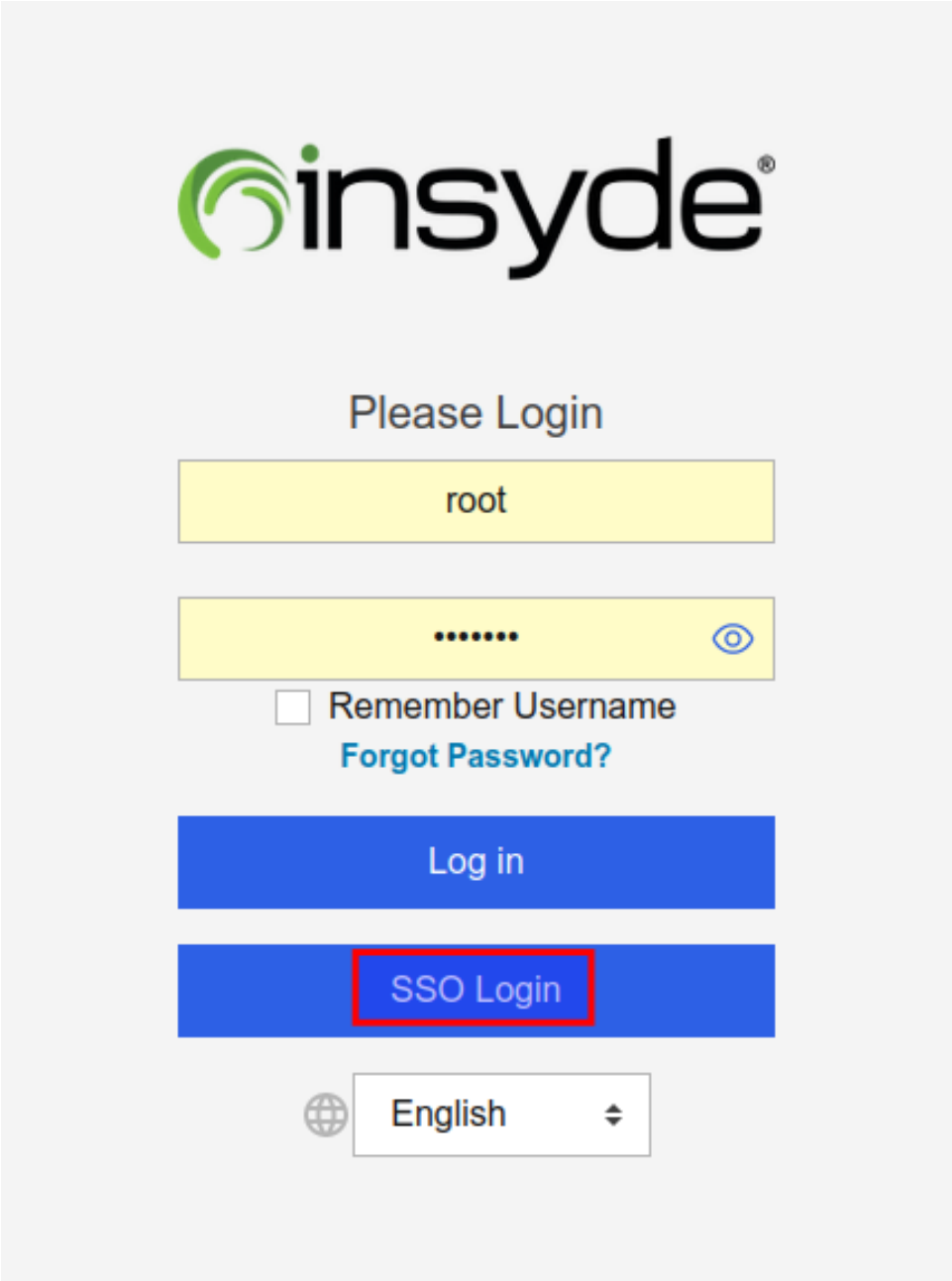
- **Network static with self-setup DNS server may cause NTP time synchronization fail**

- 

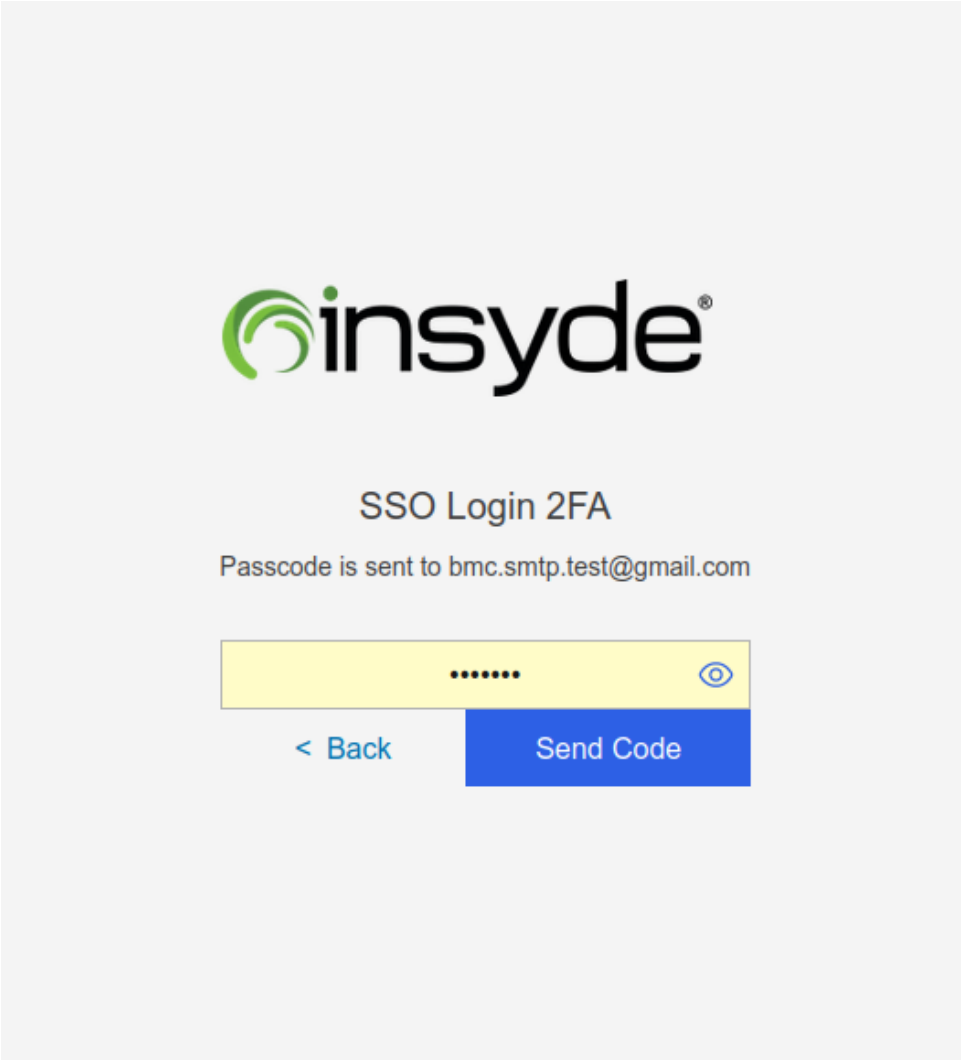
6.1.6. SSO Login

Use Firefox or Chrome to visit BMC Web login page e.g. kdc01.kerberos.com

1. Kerberos SSO login web page
   Skip name/password and directly press SSO login button



2. Kerberos SSO 2FA web page
   Will receive 2FA passcode email which is the email setup in windows 2016 server AD user(e.g. insyde) property
   e.g. bmc.smtp.test@gmail.com



- If wrong passcode 3 times, require 10 mins waiting time for new passcode validating.